

P002 — Inverted pendulum with a vertically oscillating pivot

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Release gates: 1 of 14 failed. Stamped here by `/log per skills/log/SKILL.md` step 2, which requires a failed gate to be recorded in `meta.yaml` and in a banner at the top of this document rather than quietly fixed.

`latex_clean` — fail. This document compiles with no undefined references, no undefined citations, no missing characters and no `\emph`, but seven overfull boxes exceed 10 pt: 37.1, 32.4, 27.7, 27.7, 18.6, 18.2 and 13.5 pt. Six are rows of the check-to-file table in §4, where a long script filename does not fit its cell; the seventh is a row of the provenance ledger. All are typographic, and none touches the derivation or any number in it. They are not fixed because step 3 of the log skill forbids editing the final document to clear a cosmetic gate. Thirteen further overfull boxes were cleared first by `\sloppy` and `\emergencystretch`, which are line-breaking parameters that change no text.

The other thirteen gates pass.

Abstract

A pendulum of length l carrying a point mass m , with its pivot driven vertically as $y_p(t) = A \cos \omega t$ with $A \ll l$, is stable in the inverted position when $A^2 \omega^2 > 2gl$, and then oscillates slowly about the vertical at $\Omega = \sqrt{A^2 \omega^2 / 2l^2 - g/l}$. The mechanism is that the shaking forces a fast wiggle $\xi = (A/l) \sin \Theta \cos \omega t$ whose amplitude grows with the lean, in phase with the driving term that produces it; the product of two quantities that each average to zero does not average to zero, and what survives is a restoring torque. Equivalently, the mean kinetic energy stored in the wiggle, $\frac{1}{4} m A^2 \omega^2 \sin^2 \Theta$, acts as a potential hill at the horizontal. The equation of motion is obtained from a Lagrangian and the slow dynamics by two-timescale averaging. The result is not asserted on the strength of that derivation alone: the equation of motion is checked symbolically from the geometry and against an independently formulated Newtonian constrained dynamics; the threshold is recomputed by exact Floquet theory, which contains no averaging; and the frequency is recomputed twice, once as a period measured on the exact nonlinear equation and once as a Floquet phase, both converging on the closed form at the predicted second order in A/l . Every check is also run against deliberately mutated versions of the claim, and the mutants are rejected. The averaged results carry a relative error of order $(A/l)^2$, which is measured rather than asserted.

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2. Setup

2.1 Conventions

SI units throughout. The motion is planar. The angle θ is measured from the **upward** vertical, so $\theta = 0$ is the inverted (upside-down) position and $\theta = \pi$ is the ordinary hanging position. [A] This is a convention, not a physical assumption; it costs nothing and it is chosen because the question is about the inverted position. Nothing breaks if the opposite convention is used, beyond a relabelling $\theta \rightarrow \pi - \theta$.

Averages written $\langle \cdot \rangle$ are over one drive period $T = 2\pi/\omega$, at fixed slow variables.

2.2 Symbol table

Symbol	Meaning	Units
m	mass of the bob	kg
l	length of the massless stick	m
g	gravitational acceleration, $g > 0$	m s^{-2}
A	amplitude of the pivot's vertical oscillation, $A \ll l$	m
ω	angular frequency of the pivot's oscillation	s^{-1}
$y_p(t) = A \cos \omega t$	vertical position of the pivot	m
$\theta(t)$	angle of the stick from the upward vertical	rad
$\Theta(t)$	slow part of θ	rad
$\xi(t)$	fast part of θ , $\langle \xi \rangle = 0$	rad
$\varepsilon = A/l$	small parameter	—
$\omega_0 = \sqrt{g/l}$	natural frequency of the unshaken pendulum	s^{-1}
Ω	frequency of the slow back-and-forth motion	s^{-1}
$\omega_c = \sqrt{2gl}/A$	threshold drive frequency	s^{-1}
$W(\Theta)$	effective potential per unit ml^2	s^{-2}
$U_{\text{eff}} = ml^2 W$	effective potential	J
Θ_c	angle of the barrier bounding the inverted well	rad

2.3 The problem, restated precisely

A point mass m is fixed to one end of a rigid massless stick of length l . The other end is constrained to the vertical line $x = 0$ and is driven along it with prescribed position $y_p(t) = A \cos \omega t$, with $A \ll l$. The stick may rotate freely about the pivot in a fixed vertical plane. Gravity is uniform and downward. There is no friction and no air.

Two things are asked. First, why the configuration $\theta = 0$ — which for $A = 0$ is unstable — can fail to be abandoned when ω is large. Second, the frequency of the slow oscillation about $\theta = 0$ that replaces the fall.

Note that $\theta \equiv 0$ and $\theta \equiv \pi$ are **exact** solutions of the exact equation of motion derived below, for any A and ω : with the stick vertical there is no torque about the pivot, and the bob simply translates with the pivot. [D] So the question is not whether an equilibrium exists

— two do, exactly — but whether $\theta = 0$ is stable. That distinction matters for §4, because it means the stability question can be settled exactly by Floquet theory, with no averaging, and that is what check N4 does.



3. Derivation

3.1 Coordinates and the Lagrangian

The pivot is at $(0, y_p(t))$ and the bob at

$$x = l \sin \theta, \quad y = y_p + l \cos \theta. \quad (1)$$

Differentiating,

$$\dot{x} = l \dot{\theta} \cos \theta, \quad (2)$$

$$\dot{y} = \dot{y}_p - l \dot{\theta} \sin \theta, \quad \dot{y}_p = -A \omega \sin \omega t. \quad (3)$$

The kinetic energy is $T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$. Writing the two squares out,

$$\dot{x}^2 = l^2 \dot{\theta}^2 \cos^2 \theta, \quad (4)$$

$$\dot{y}^2 = \dot{y}_p^2 - 2l \dot{y}_p \dot{\theta} \sin \theta + l^2 \dot{\theta}^2 \sin^2 \theta, \quad (5)$$

and adding them, the two $l^2 \dot{\theta}^2$ pieces combine through $\cos^2 \theta + \sin^2 \theta = 1$:

$$\dot{x}^2 + \dot{y}^2 = l^2 \dot{\theta}^2 + \dot{y}_p^2 - 2l \dot{y}_p \dot{\theta} \sin \theta, \quad (6)$$

so that

$$T = \frac{1}{2}ml^2 \dot{\theta}^2 + \frac{1}{2}m\dot{y}_p^2 - ml \dot{y}_p \dot{\theta} \sin \theta. \quad (7)$$

The potential energy is

$$V = mgy = mg y_p + mgl \cos \theta, \quad (8)$$

and therefore

$$L = T - V = \frac{1}{2}ml^2 \dot{\theta}^2 - ml \dot{y}_p \dot{\theta} \sin \theta - mgl \cos \theta + \underbrace{\frac{1}{2}m\dot{y}_p^2 - mg y_p}_{\text{function of } t \text{ alone}}. \quad (9)$$

The bracketed group depends on t only. Both $\partial/\partial\theta$ and $\partial/\partial\dot{\theta}$ annihilate it, so it cannot appear anywhere in the Euler–Lagrange equation for θ and may be dropped. (It is not necessary to invoke the weaker “total time derivative” argument; a function of t alone is killed directly.) Keep [D]

$$\tilde{L} = \frac{1}{2}ml^2 \dot{\theta}^2 - ml \dot{y}_p \dot{\theta} \sin \theta - mgl \cos \theta. \quad (10)$$

3.2 Equation of motion

$$\frac{\partial \tilde{L}}{\partial \dot{\theta}} = ml^2 \dot{\theta} - ml \dot{y}_p \sin \theta, \quad (11)$$

$$\frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{\theta}} = ml^2 \ddot{\theta} - ml \ddot{y}_p \sin \theta - ml \dot{y}_p \dot{\theta} \cos \theta, \quad (12)$$

$$\frac{\partial \tilde{L}}{\partial \theta} = -ml \dot{y}_p \dot{\theta} \cos \theta + mgl \sin \theta. \quad (13)$$

Subtracting (13) from (12) and setting the result to zero,

$$ml^2 \ddot{\theta} - ml \ddot{y}_p \sin \theta - ml \dot{y}_p \dot{\theta} \cos \theta + ml \dot{y}_p \dot{\theta} \cos \theta - mgl \sin \theta = 0. \quad (14)$$

The two terms in $\dot{y}_p \dot{\theta} \cos \theta$ cancel identically. This cancellation is worth pausing on: it is why the pivot's velocity drops out entirely and only its acceleration survives. Dividing by ml , [D]

$$l \ddot{\theta} = (g + \ddot{y}_p) \sin \theta. \quad (15)$$

With $\ddot{y}_p = -A\omega^2 \cos \omega t$,

$$\boxed{l \ddot{\theta} = (g - A\omega^2 \cos \omega t) \sin \theta} \quad (16)$$

Equation (15) says exactly what the equivalence principle would say: in the frame of the accelerating pivot the bob feels an effective gravitational acceleration $g_{\text{eff}}(t) = g + \ddot{y}_p$, and the pendulum equation is the usual one with $g \rightarrow g_{\text{eff}}$. This is a consistency remark, not an independent derivation; the independent derivation is the Newtonian one in check N2, which does not pass through a Lagrangian at all. [D]

Setting $A = 0$ in (16) gives $l \ddot{\theta} = g \sin \theta$, the rigid pendulum measured from the top, whose linearization $\ddot{\theta} = \omega_0^2 \theta$ grows like $e^{\omega_0 t}$. [D] That is the fall the problem asks us to explain away.

It is convenient to divide (16) by l and introduce $\varepsilon = A/l$ and $\omega_0^2 = g/l$:

$$\ddot{\theta} = (\omega_0^2 - \varepsilon \omega^2 \cos \omega t) \sin \theta. \quad (17)$$

3.3 Separation of the fast and slow motions

Assumption 1 (two timescales). Write [A]

$$\theta(t) = \Theta(t) + \xi(t), \quad \langle \xi \rangle = 0, \quad (18)$$

where ξ oscillates at the drive frequency ω with amplitude $O(\varepsilon)$, and Θ varies only on a timescale long compared with $T = 2\pi/\omega$. What this buys: it converts a non-autonomous equation into an autonomous one for Θ . What breaks without it: if the two timescales are not separated — if Ω is comparable to ω , or if ξ is not small — then ξ cannot be solved for at leading order independently of Θ and the whole construction collapses. §3.6 shows that the separation is a consequence of $A \ll l$ rather than an extra hypothesis, and check N3 measures the error it commits.

Substituting (18) into (17) and expanding the sine,

$$\sin(\Theta + \xi) = \sin \Theta \cos \xi + \cos \Theta \sin \xi = \sin \Theta + \xi \cos \Theta + O(\xi^2), \quad (19)$$

so

$$\ddot{\Theta} + \ddot{\xi} = (\omega_0^2 - \varepsilon \omega^2 \cos \omega t) (\sin \Theta + \xi \cos \Theta) + O(\xi^2). \quad (20)$$

Assumption 2 (truncation). Terms of $O(\xi^2) = O(\varepsilon^2)$ inside the sine are dropped. What it buys: (20) becomes linear in ξ . What breaks: the retained slow equation is then accurate only to leading order; the neglected terms are of relative size ε^2 , and check N3 measures exactly that and finds the predicted second-order convergence. [A]

Before splitting (20), count the sizes of the terms, since the split is the one place where knowing the answer could do the work instead of the argument.

To do the counting honestly one number is needed: how large ω_0^2 is compared with $\varepsilon^2\omega^2$. Define

$$\mu \equiv \frac{2gl}{A^2\omega^2} = \frac{2\omega_0^2}{\varepsilon^2\omega^2}, \quad \text{so that} \quad \omega_0^2 = \frac{1}{2}\mu\varepsilon^2\omega^2. \quad (20a)$$

μ is not a free parameter dragged in for convenience: it is the same combination that will turn up in (34) as $\cos\Theta_c$, and the stability condition derived below is exactly $\mu < 1$. So in the whole region where the inverted position is held, $0 < \mu \leq 1$, with $\mu = 1$ at threshold and $\mu \rightarrow 0$ far above it. Every statement in the table below is therefore made for $\mu \in (0, 1]$ and not for some unstated “large ω ” regime. [D]

All the angular factors — $\sin\Theta$, $\sin\Theta\cos\Theta$, Θ — are $O(1)$ and are not part of the counting; what is being compared is the coefficients in front of them.

term	coefficient	block
$\varepsilon\omega^2 \cos\omega t \sin\Theta$	$\varepsilon\omega^2$	fast
$\ddot{\xi}$	$\sim \omega^2\xi \sim \varepsilon\omega^2$	fast
$\varepsilon\omega^2 \cos(\omega t) \xi \cos\Theta$	$\varepsilon^2\omega^2$	slow
$\ddot{\Theta}$	$\sim \Omega^2 = \frac{1}{2}\varepsilon^2\omega^2(1 - \mu)$	slow
$\omega_0^2 \sin\Theta$	$\omega_0^2 = \frac{1}{2}\mu\varepsilon^2\omega^2$	slow

There are two distinct blocks, separated by one factor of ε and by nothing else: everything in the first block is $O(\varepsilon\omega^2)$ and everything in the second is $O(\varepsilon^2\omega^2)$. The smallness of ε is the only smallness used anywhere in this derivation. [D]

Within the slow block, all three terms are of the same order and none may be dropped. In particular ω_0^2 is not negligible against $\varepsilon^2\omega^2$: by (20a) it is exactly half of it at threshold, and at most half of it anywhere in the stable region. Far above threshold, $\mu \ll 1$, it becomes subdominant within its own block, but it never leaves that block and is never dropped — and it is not dropped anywhere below. This matters more than it looks: if $\omega_0^2 \sin\Theta$ were discarded from the slow equation, the averaged dynamics would lose gravity altogether, $\Theta = 0$ would be stable for every ω , and there would be no threshold to derive. The existence of a threshold is the statement that these two slow terms are comparable. [D]

One degeneracy is worth flagging now rather than meeting it later. At $\mu = 1$ the coefficient of $\ddot{\Theta}$ vanishes, $\Omega \rightarrow 0$, and the two surviving slow terms cancel at leading order in Θ . The ordering inside the slow block degenerates there and the quartic term takes over; that is the marginal case, and it is treated separately in §3.8.

Nothing in this counting is a choice. The first block must balance against itself because it is the largest thing present; the second must balance against itself because there is nothing else left at that order.

3.4 The fast equation

At order $\varepsilon\omega^2$, (20) gives

$$\ddot{\xi} = -\varepsilon\omega^2 \cos(\omega t) \sin\Theta. \quad (21)$$

[A]

Assumption 3. Θ is held fixed while (21) is integrated over one drive period, which is legitimate precisely because Θ is slow. Integrating twice,

$$\dot{\xi} = -\varepsilon\omega \sin \Theta \sin(\omega t) + c_1, \quad (22)$$

$$\xi = \varepsilon \sin \Theta \cos(\omega t) + c_1 t + c_2. \quad (23)$$

$c_1 = 0$: a nonzero c_1 produces a term growing linearly in t , which contradicts the assumption [D] that ξ stays $O(\varepsilon)$. $c_2 = 0$: a constant in ξ is by definition part of the slow motion and has already been named Θ , and it is excluded by $\langle \xi \rangle = 0$. Hence

$$\xi = \varepsilon \sin \Theta \cos \omega t = \frac{A}{l} \sin \Theta \cos \omega t. \quad (24)$$

Two things are worth noticing about (24), because they are the mechanism of the whole effect. First, the two powers of ω brought down by $\ddot{y}_p$ are removed exactly by the double integration, so the amplitude of the fast wiggle is $\varepsilon \sin \Theta$, independent of ω . Second, that amplitude is proportional to $\sin \Theta$: the further the stick leans, the harder the shaking rattles it. [D] Check N6 measures both statements directly.

3.5 Averaging, and the effective potential

Average (20) over one drive period at fixed $\Theta, \dot{\Theta}$. Term by term:

$$\langle \ddot{\xi} \rangle = \frac{1}{T} [\dot{\xi}]_0^T = 0 \quad \text{since } \dot{\xi} \text{ is } T\text{-periodic}, \quad (25)$$

$$\langle \cos \omega t \rangle = 0, \quad \langle \xi \rangle = 0, \quad (26)$$

$$\langle \cos(\omega t) \xi \rangle = \varepsilon \sin \Theta \langle \cos^2 \omega t \rangle = \frac{1}{2} \varepsilon \sin \Theta. \quad (27)$$

Equation (27) is the crux. The driving term in (17) is proportional to $\cos \omega t$, and the wiggle [D] it produces is also proportional to $\cos \omega t$ — in phase, because (21) is a pure second derivative with no damping. A product of two quantities each averaging to zero need not average to zero, and here it does not: $\langle \cos^2 \rangle = \frac{1}{2}$. The factor $\frac{1}{2}$ in the final answer comes from here and from nowhere else, and it is not adjustable.

Averaging (20) and using (25)–(27),

$$\ddot{\Theta} = \omega_0^2 \sin \Theta - \varepsilon \omega^2 \cdot \frac{1}{2} \varepsilon \sin \Theta \cos \Theta,$$

that is

$$\boxed{\ddot{\Theta} = \frac{g}{l} \sin \Theta - \frac{A^2 \omega^2}{2l^2} \sin \Theta \cos \Theta} \quad (28)$$

Write (28) as $\ddot{\Theta} = -dW/d\Theta$. Using $\sin \Theta \cos \Theta = \frac{1}{2} \sin 2\Theta$ and $\sin^2 \Theta = \frac{1}{2}(1 - \cos 2\Theta)$,

$$W(\Theta) = \frac{g}{l} \cos \Theta + \frac{A^2 \omega^2}{4l^2} \sin^2 \Theta. \quad (29)$$

Verifying by differentiation, step by step:

$$\frac{dW}{d\Theta} = -\frac{g}{l} \sin \Theta + \frac{A^2 \omega^2}{4l^2} \cdot 2 \sin \Theta \cos \Theta = -\frac{g}{l} \sin \Theta + \frac{A^2 \omega^2}{2l^2} \sin \Theta \cos \Theta, \quad (30)$$

so $-dW/d\Theta$ reproduces the right-hand side of (28) exactly. Multiplying by ml^2 to get an energy,

$$U_{\text{eff}}(\Theta) = mgl \cos \Theta + \frac{mA^2\omega^2}{4} \sin^2 \Theta. \quad (31)$$

The added term is the time-averaged kinetic energy of the fast motion. From (24), $\dot{\xi} = -\varepsilon\omega \sin \Theta \sin \omega t$, so the kinetic energy carried by the fast part of the angular motion is

$$\left\langle \frac{1}{2} ml^2 \dot{\xi}^2 \right\rangle = \frac{1}{2} ml^2 \varepsilon^2 \omega^2 \sin^2 \Theta \langle \sin^2 \omega t \rangle = \frac{mA^2\omega^2}{4} \sin^2 \Theta, \quad (32)$$

identical to the second term of (31). This is an internal consistency check of the algebra that costs nothing and would have caught a wrong factor in (27): the mean kinetic energy stored in the jitter acts as a potential for the slow motion, largest where the jitter is largest, at $\Theta = \pi/2$. That hill at the horizontal is what holds the pendulum up.

3.6 Equilibria, stability, and the slow frequency

From (30), the stationary points of W satisfy

$$\sin \Theta \left[-\frac{g}{l} + \frac{A^2\omega^2}{2l^2} \cos \Theta \right] = 0, \quad (33)$$

giving $\Theta = 0$, $\Theta = \pi$, and, when the bracket can vanish,

$$\cos \Theta_c = \frac{2gl}{A^2\omega^2}, \quad \text{which has a solution iff } A^2\omega^2 \geq 2gl. \quad (34)$$

Differentiating (30) once more,

$$\frac{d^2W}{d\Theta^2} = -\frac{g}{l} \cos \Theta + \frac{A^2\omega^2}{2l^2} \cos 2\Theta. \quad (35)$$

At the inverted position,

$$W''(0) = -\frac{g}{l} + \frac{A^2\omega^2}{2l^2}, \quad (36)$$

so $\Theta = 0$ is a minimum of W , and hence stable for the slow dynamics, precisely when

$$\boxed{A^2\omega^2 > 2gl \iff \omega > \omega_c \equiv \frac{\sqrt{2gl}}{A} \iff \frac{A\omega}{l} > \sqrt{2}\omega_0} \quad (37)$$

At the hanging position, $\cos \pi = -1$ and $\cos 2\pi = 1$, so

$$W''(\pi) = \frac{g}{l} + \frac{A^2\omega^2}{2l^2} > 0 \quad \text{always}, \quad (38)$$

i.e. shaking the pivot never destabilizes the hanging position at this order, and stiffens it: the slow frequency about $\Theta = \pi$ is $\sqrt{g/l + A^2\omega^2/2l^2}$, which reduces to ω_0 as $A \rightarrow 0$. That reduction is one of the self-checks in §4.6. [D]

At Θ_c , writing $q \equiv \cos \Theta_c = 2gl/(A^2\omega^2)$ and using $g/l = \frac{1}{2}q A^2\omega^2/l^2$ from (34),

$$W''(\Theta_c) = -\frac{g}{l}q + \frac{A^2\omega^2}{2l^2}(2q^2 - 1) = \frac{A^2\omega^2}{2l^2}(-q^2 + 2q^2 - 1) = \frac{A^2\omega^2}{2l^2}(q^2 - 1) < 0 \quad (39)$$

for $q < 1$. [D] So Θ_c is a maximum of W : it is the top of the barrier separating the inverted well from the fall toward $\Theta = \pi$. The “back and forth motion” of the problem statement is motion inside that well, and it exists only for $|\Theta| < \Theta_c$.

Finally, for small departures from the inverted position, expand (28) with $\sin \Theta \approx \Theta$, $\cos \Theta \approx 1$:

$$\ddot{\Theta} = \left(\frac{g}{l} - \frac{A^2 \omega^2}{2l^2} \right) \Theta = -W''(0) \Theta, \quad (40)$$

which is simple harmonic with

$$\Omega = \sqrt{\frac{A^2 \omega^2}{2l^2} - \frac{g}{l}} = \sqrt{\frac{A^2 \omega^2 - 2gl}{2l^2}} \quad (41)$$

and, when $A\omega \gg \sqrt{gl}$,

$$\Omega \simeq \frac{A\omega}{\sqrt{2}l}. \quad (42)$$

3.7 Is the timescale separation self-consistent?

This has to be checked rather than assumed, because Assumption 1 was used to get (41) and (41) is what tells us whether Assumption 1 holds. From (41),

$$\frac{\Omega}{\omega} = \sqrt{\frac{\varepsilon^2}{2} - \frac{\omega_0^2}{\omega^2}} < \frac{\varepsilon}{\sqrt{2}}. \quad (43)$$

So the slow frequency is smaller than the drive frequency by at least $\varepsilon/\sqrt{2}$, for every parameter set where the inverted position is stable at all. The separation of timescales is therefore a consequence of $A \ll l$, not an independent assumption — a point worth making because it is easy to present it as one and thereby hide a circularity. Similarly, (37) forces [D]

$$\omega > \frac{\sqrt{2}\omega_0}{\varepsilon} \gg \omega_0, \quad (44)$$

so “ ω large enough” in the problem statement is not vague: stability itself requires the drive to be fast compared with the pendulum’s own natural frequency, by a factor at least $\sqrt{2}/\varepsilon$.

3.8 Behaviour exactly at threshold

Expanding (29) to fourth order, with $\cos \Theta = 1 - \Theta^2/2 + \Theta^4/24$ and $\sin^2 \Theta = \Theta^2 - \Theta^4/3$,

$$W = \omega_0^2 + \frac{1}{2}\Theta^2 \left(\frac{\varepsilon^2 \omega^2}{2} - \omega_0^2 \right) + \Theta^4 \left(\frac{\omega_0^2}{24} - \frac{\varepsilon^2 \omega^2}{12} \right) + O(\Theta^6) \equiv \omega_0^2 + \frac{1}{2}\Omega^2 \Theta^2 + \beta \Theta^4, \quad (45)$$

$$\beta = \frac{\omega_0^2}{24} - \frac{\varepsilon^2 \omega^2}{12} = -\left(\frac{\Omega^2}{6} + \frac{\omega_0^2}{8} \right), \quad (46)$$

where the second form uses $\varepsilon^2 \omega^2 = 2(\Omega^2 + \omega_0^2)$ from (41). [D] $\beta < 0$ always. At threshold $\Omega = 0$ and $\beta = -\omega_0^2/8 < 0$, so W has a quartic maximum at $\Theta = 0$: exactly at $A^2 \omega^2 = 2gl$ the inverted position is unstable, and the inequality in (37) is strict. This also means the effective well is anharmonic in the softening direction — the slow period lengthens with amplitude — which §4.4 uses quantitatively.

3.9 The amplitude–frequency relation for the effective well

New in round 2, in response to finding F1.2. In round 1 the amplitude dependence of the slow period was quoted from recall and tagged [R]. It is short enough to derive, so it is derived here and the recall is removed from the chain. The procedure is the standard Lindstedt–Poincaré expansion, but nothing below rests on recalling it: the calculation is written out in full, and its result is tested directly in check N7.

From (45)–(46), differentiating the quartic truncation of W ,

$$\ddot{\Theta} = -\frac{dW}{d\Theta} = -\Omega^2\Theta - 4\beta\Theta^3 + O(\Theta^5). \quad (47)$$

Write $\alpha \equiv 4\beta$, so that the slow motion near the bottom of the well obeys a Duffing equation,

$$\ddot{\Theta} + \Omega^2\Theta + \alpha\Theta^3 = 0, \quad \Theta(0) = a, \quad \dot{\Theta}(0) = 0. \quad (48)$$

The expansion parameter is $\alpha a^2/\Omega^2$, assumed small. What it buys: a closed-form amplitude correction. What breaks without it: nothing in the main result, since (48) is only used to interpret a residual in check N3; but the relation itself fails as $a \rightarrow \Theta_c$, where the quartic truncation of W is no longer the whole story. [A]

A naive expansion in α at fixed t produces terms growing like t , so instead stretch time by the unknown true frequency $\tilde{\omega}$, putting $\tau = \tilde{\omega}t$ so that the solution is 2π -periodic in τ by construction. With $' = d/d\tau$, (48) becomes

$$\tilde{\omega}^2\Theta'' + \Omega^2\Theta + \alpha\Theta^3 = 0. \quad (49)$$

Expand both the solution and the frequency,

$$\Theta = \Theta_0 + \Theta_1 + \cdots, \quad \tilde{\omega}^2 = \Omega^2 + c_1 + \cdots, \quad (50)$$

with Θ_1 and c_1 first order in α . At zeroth order (49) reads $\Omega^2(\Theta_0'' + \Theta_0) = 0$, which with $\Theta_0(0) = a$, $\Theta_0'(0) = 0$ gives

$$\Theta_0 = a \cos \tau. \quad (51)$$

Collecting the terms of first order in α in (49),

$$\Omega^2\Theta_1'' + c_1\Theta_0'' + \Omega^2\Theta_1 + \alpha\Theta_0^3 = 0,$$

and using $\Theta_0'' = -a \cos \tau$ from (51), together with the triple-angle identity $\cos^3 \tau = \frac{3}{4} \cos \tau + \frac{1}{4} \cos 3\tau$ (which follows from $\cos 3\tau = 4 \cos^3 \tau - 3 \cos \tau$),

$$\Omega^2(\Theta_1'' + \Theta_1) = c_1 a \cos \tau - \alpha a^3 \cos^3 \tau = \left(c_1 a - \frac{3}{4} \alpha a^3\right) \cos \tau - \frac{1}{4} \alpha a^3 \cos 3\tau. \quad (52)$$

The operator on the left annihilates $\cos \tau$, so a $\cos \tau$ term on the right is resonant and drives a response growing like $\tau \sin \tau$. That contradicts the periodicity imposed by the stretching, so its coefficient must vanish: [D]

$$c_1 = \frac{3}{4} \alpha a^2. \quad (53)$$

Hence $\tilde{\omega}^2 = \Omega^2 + \frac{3}{4} \alpha a^2$, and taking the square root and expanding,

$$\tilde{\omega} = \Omega \sqrt{1 + \frac{3\alpha a^2}{4\Omega^2}} = \Omega \left(1 + \frac{3\alpha a^2}{8\Omega^2}\right) + O(\alpha^2 a^4). \quad (54)$$

The period is $T(a) = 2\pi/\tilde{\omega}$, so the relative period shift is

$$\frac{T(a)}{T(0)} - 1 = -\frac{3\alpha a^2}{8\Omega^2} + O(a^4) = -\frac{3\beta a^2}{2\Omega^2} + O(a^4), \quad (55)$$

using $\alpha = 4\beta$. Substituting $\beta = -(\Omega^2/6 + \omega_0^2/8)$ from (46) and simplifying,

$$\frac{T(a)}{T(0)} - 1 = \frac{3a^2}{2\Omega^2} \left(\frac{\Omega^2}{6} + \frac{\omega_0^2}{8}\right) = a^2 \left(\frac{1}{4} + \frac{3\omega_0^2}{16\Omega^2}\right). \quad (56)$$

Both terms are positive, so the slow period always lengthens with amplitude: the effective well is softening, everywhere in the stable region. That is the sign already visible in figure 3, where the well is flatter than a parabola at its edges, and it is the sign $\beta < 0$ from (46). Equation (56) is what check N3's column (c) measures inside the pendulum, and check N7 measures directly on (48) with no pendulum present. [D]

4. Verification

Eight computational checks, plus the self-check battery. Every script is standalone, runs from a clean interpreter, pins its library versions in a comment, and fixes a seed where randomness is used. Library versions as run:

```
python 3.11.15, sympy 1.14.0, numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.9
```

Checks do not map one-to-one onto files, so here is the correspondence. All files are in `assets/code/` and all are listed verbatim in the appendix.

check	what it tests	file	output	section
N1	equation of motion, symbolically from the geometry	<code>check1_eom_symbolic.py</code>	<code>out_check1.txt</code>	§4.1
N2	equation of motion against Newton with a constraint force	<code>check2_newton_vs_lagrange.py</code>	<code>out_check2.txt</code>	§4.2
—	diagnosis of N2's failing set	<code>diag_set8.py</code>	quoted inline	§4.2
N3	slow period from the exact nonlinear equation	<code>check3_slow_frequency.py</code>	<code>out_check3.txt</code>	§4.3
N4	stability threshold by exact Floquet theory	<code>check4_floquet_boundary.py</code>	<code>out_check4.txt</code>	§4.4
N5, N6	barrier angle; fast-ripple envelope	<code>check5_basin_and_ripple.py</code>	<code>out_check5.txt</code>	§4.5
N7	amplitude–frequency relation, on the bare Duffing equation	<code>check6_duffing_shiftout.py</code>	<code>out_check6.txt</code>	§4.8
N8	slow frequency from the Floquet phase	<code>check7_floquet_frequency.py</code>	<code>out_check7.txt</code>	§4.9

N7 and N8 are new in round 2. They are placed at the end of §4 rather than in check order so that the section numbers referenced from the frozen sections of §3 keep pointing where they did in round 1.

Every check was re-run from scratch for this round. Checks N1–N6 reproduced their round-1 output byte for byte, including the failure of N2’s set 8, which is what a seeded, pinned, deterministic script is supposed to do. Outputs below are verbatim; nothing is paraphrased, abridged, or carried over.

A general remark on how these checks are built, since it is the part most worth attacking. A passing check on a claim I already believe is weak evidence. Every check below therefore also runs against **deliberately mutated** versions of the claim — a flipped sign, a missing factor of two, a dropped $\sin \Theta$ — and reports whether the mutant is rejected. A check that cannot reject a mutant is reported as vacuous rather than as a pass. Where the underlying claim is asymptotic, the check is a **convergence test** rather than a fixed-tolerance test: the residual is required to vanish at the predicted order in ε , which is a much harder target to hit by accident than a single tolerance.

4.1 Check N1 — the equation of motion, symbolically, from the geometry

What it tests: that (16) follows from (1)–(10). The script builds $x(t)$ and $y(t)$ from the geometry, forms T and V from their definitions, applies the Euler–Lagrange operator through sympy, solves for $\ddot{\theta}$, and subtracts the claim. No expression from §3.2 is reused. [N]

Output:

```
L = m*(-2*g*(A*cos(omega*t) + l*cos(theta(t))) + l**2*cos(theta(t))*2*
    Derivative(theta(t), t)**2 + (A*omega*sin(omega*t) + l*sin(theta(t))*
    Derivative(theta(t), t))**2)/2
Euler-Lagrange expression (should equal 0):
l*m*(A*omega**2*sin(theta(t))*cos(omega*t) - g*sin(theta(t)) + l*Derivative(
    theta(t), (t, 2)))
theta'' solved from Euler-Lagrange:
(-A*omega**2*cos(omega*t) + g)*sin(theta(t))/l
claimed theta'' = (-A*omega**2*cos(omega*t) + g)*sin(theta(t))/l
residual (solved - claimed) = 0
CHECK N1: PASS
residual against sign-flipped mutant = -2*A*omega**2*sin(theta(t))*cos(omega*t)
    /l
MUTANT REJECTED: yes
A -> 0 limit of theta'' : g*sin(theta(t))/l
```

What this would have caught: any algebra slip between (1) and (16) — a dropped cross term in (6), a sign error in (12) or (13), a failure of the $\dot{y}_p \dot{\theta} \cos \theta$ cancellation. The sign-flipped mutant is rejected with a nonzero residual, so the test is not vacuous.

What it would **not** have caught: a wrong setup. If (1) itself encoded the geometry incorrectly — the pivot moving horizontally, say, or θ measured from the wrong axis — sympy would faithfully derive the equation of motion for that wrong problem and report PASS. It also says nothing about §3.3 onwards. Check N2 addresses the first gap; checks N3–N5 address the second.

4.2 Check N2 — the equation of motion against Newton with a constraint force

What it tests: (16) against a second formulation of the same mechanics that never forms a Lagrangian. Newton’s second law is written in Cartesian components for the bob, the rigid stick is represented by a force of unknown magnitude directed along it, and that magnitude is eliminated by differentiating $|\mathbf{r} - \mathbf{p}(t)|^2 = l^2$ twice. Both systems are started from the same physical state and integrated with DOP853; the angles are compared. [N]

This check failed on its first run, and the failure is reported here before the revision. The first version applied one criterion to all eight parameter sets: $\max |\Delta\theta| < 10^{-6}$ rad over a 6 s window at $\text{rtol}=1\text{e-}11$. Set 8 failed it, with $\max |\Delta\theta| = 4.4 \times 10^{-1}$. Set 8 has $\omega/\omega_c = 0.47$, i.e. below threshold, so the pendulum falls and then whirls, and in that regime the motion has a positive Lyapunov exponent: two numerically exact but differently discretized integrations of the same equations separate exponentially. The diagnostic run (`diag_set8.py`, also in `assets/code/`) gave

<code>rtol=1e-09</code>	<code>max dth =6.167e+00</code>	<code>max drift=5.794e-01</code>	<code>separation time=2.207 s</code>	<code>(</code>
<code>tmax=6.00)</code>				
<code>rtol=1e-11</code>	<code>max dth =4.432e-01</code>	<code>max drift=5.803e-01</code>	<code>separation time=3.385 s</code>	<code>(</code>
<code>tmax=6.00)</code>				
<code>rtol=1e-13</code>	<code>max dth =1.301e-02</code>	<code>max drift=1.453e-02</code>	<code>separation time=4.068 s</code>	<code>(</code>
<code>tmax=6.00)</code>				
<code>rtol=1e-14</code>	<code>max dth =9.992e-04</code>	<code>max drift=1.104e-03</code>	<code>separation time=4.341 s</code>	<code>(</code>
<code>tmax=6.00)</code>				

The discrepancy falls by four orders of magnitude as the tolerance is tightened, in lockstep with the independent witness $||\mathbf{u}| - l|/l$, which the θ -equation cannot drift in by construction, and the time at which the two curves separate moves steadily later. That is integrator error, not a disagreement between the two formulations.

What I changed, and it was the check and not the derivation. The fixed-tolerance criterion is kept and its verdict reported unchanged; any set that fails it is then put through a tolerance-refinement study, and a discrepancy that falls with rtol is attributed to integration rather than to physics. Nothing in §3 was touched. The cost of this is stated in the coverage note below.

Output of the revised check (the eight-set table reproduces the first version's numbers exactly, including the failure):

```
CHECK N2 -- scalar theta-equation vs Newton-with-constraint-force.
Both integrations start from the same physical state.
Primary criterion: max|dthet| < 1e-6 rad over the window at rtol=1e-11.
A real disagreement would show as a discrepancy that does NOT fall when
rtol is tightened. Both integrators are DOP853, so a shared
discretisation bias is not credible at these tolerances.
```

set	l	A	omega	w/w_c	th0	tmax	max dth	u drift
	verdict							
1	0.479	0.01001	486.2	1.59	0.0274	2.58	1.650e-11	2.07e-13
	PASS							
2	0.461	0.01463	495.2	2.41	-0.1346	2.54	1.062e-10	2.14e-12
	PASS							
3	0.897	0.01741	504.8	2.10	-0.2725	2.49	3.538e-12	5.16e-12
	PASS							
4	0.913	0.03391	247.9	1.99	0.0991	5.07	3.221e-11	1.18e-12
	PASS							
5	0.775	0.01829	364.6	1.71	0.0247	3.45	8.830e-12	2.63e-13
	PASS							
6	1.318	0.03133	250.4	1.54	-0.1594	5.02	1.563e-11	1.71e-12
	PASS							
7	0.752	0.01472	183.4	0.70	0.2748	6.00	2.214e-08	7.32e-10
	PASS							
8	0.256	0.00886	119.7	0.47	0.0535	6.00	4.432e-01	5.80e-01
	FAIL							

```
Primary criterion: 7/8 sets passed.

Tolerance-refinement study on the set(s) that failed.
If the discrepancy is integrator error it falls with rtol, and the
constraint residual |u|-l falls with it. If the two formulations
actually disagree, neither falls.
  set 8:  l=0.2564  A=0.00886  omega=119.7  th0=0.0535
          rtol=1e-08  max|dth|=1.033e+01  max||u|-l|/l=6.716e-01
```

```

rtol=1e-10    max|dth|=1.218e+00    max||u|-1|/1=6.359e-01    (x1.18e-01)
rtol=1e-12    max|dth|=8.278e-02    max||u|-1|/1=9.836e-02    (x6.80e-02)
rtol=1e-13    max|dth|=1.301e-02    max||u|-1|/1=1.453e-02    (x1.57e-01)
first 20 drive periods (t < 1.0499 s): max|dth|=1.325e-10    -> PASS

Mutation test (super-threshold set, where the primary criterion holds):
claim as derived          : max|dtheta| = 6.495e-12
driving term sign flipped : max|dtheta| = 1.597e-02
MUTANT REJECTED: yes

```

What this would have caught: a wrong equation of motion, including the class of error N1 cannot see, because the Newtonian route rebuilds the dynamics from force balance and a geometric constraint rather than from $T - V$. The sign-flipped mutant separates by 1.6×10^{-2} rad against 6.5×10^{-12} for the claim, a ratio of 10^9 , so the check is not vacuous. It would also have caught a wrong pivot acceleration, since $\ddot{y}_p$ enters both formulations independently.

What it would **not** have caught: a shared error in the geometry, which both formulations inherit from (1) — if the pivot were supposed to move horizontally, both would agree on the wrong answer. It would not catch anything about §3.3 onwards. And, as the failure showed, it certifies agreement only over times short compared with the Lyapunov time in the whirling regime; in that regime the check verifies the equations agree for the first tens of drive periods (1.3×10^{-10} rad over 20 drive periods for set 8) and nothing beyond. For the regime the problem is actually about — near-inverted, above threshold — agreement holds over the whole window at the 10^{-11} level.

4.3 Check N3 — the slow frequency, measured from the exact equation

What it tests: equation (41), the headline result. The exact equation (16) is integrated — no averaging anywhere inside the integrator — and the period of the slow motion is read off from a straight-line fit to its zero-crossing times. Ω enters only at the end, as the number being compared against. [N]

The design point is that a single tolerance would prove little, so Ω is held fixed at 6.000 s^{-1} while $\varepsilon = A/l$ is swept down by a factor of 16, with ω solved for at each step. The residual must then fall like ε^2 , the order of the first neglected term. Three quantities are reported so that the two independent sources of discrepancy can be separated: (a) measured period against the harmonic prediction $2\pi/\Omega$; (b) measured period against the period of the averaged equation (28) at the same amplitude, which isolates the error of the averaging; (c) the averaged equation's own period against $2\pi/\Omega$, which is the anharmonic shift of §3.8 and is not an error at all. [D] The analytic size of (c) is eq. (56), derived in §3.9, and is printed for comparison. In round 1 that relation was recalled rather than derived; it is now derived, and check N7 in §4.8 tests it on its own.

Output:

```

CHECK N3 -- slow frequency measured from the exact equation of motion.
g = 9.81, l = 1.0, theta(0) = 0.02 rad, thetadot(0) = 0.
Omega is held at 6.000 rad/s while eps = A/l is swept; omega is solved
for from the claim at each eps. If the claim is right the measured
period converges on 2*pi/Omega like eps^2. A residual that plateaus,
or that falls at the wrong order, is a failure.

Predicted anharmonic period shift at theta(0) = 0.02: +1.2044e-04 (relative).
This is an eps-independent floor under
comparison (a) and must NOT be read as an error of the averaging.

eps=A/l      A      omega      w/Om      T_meas      (a)=T/Th-1      (b)=T/Tav-1      (
c)=Tav/Th-1  (b)/eps^2  ncross      fitres
0.10000      0.10000      95.72      16.0      1.039845      -7.021e-03      -7.141e-03
+1.205e-04      0.714      24      1.38e-03

```

0.05000	0.05000	191.44	31.9	1.045450	-1.669e-03	-1.789e-03
	+1.205e-04	0.716	24	3.53e-04		
0.02500	0.02500	382.87	63.8	1.046845	-3.365e-04	-4.569e-04
	+1.205e-04	0.731	24	9.06e-05		
0.01250	0.01250	765.75	127.6	1.047203	+4.747e-06	-1.157e-04
	+1.205e-04	0.740	24	2.30e-05		
0.00625	0.00625	1531.49	255.2	1.047293	+9.086e-05	-2.959e-05
	+1.205e-04	0.758	24	5.75e-06		
Decomposition check: (a) should equal (b) + (c) + 0(second order).						
eps= 0.10000:	(a)=-7.0214e-03	(b)+(c)=-7.0205e-03	residual=-8.60e-07			
eps= 0.05000:	(a)=-1.6689e-03	(b)+(c)=-1.6687e-03	residual=-2.16e-07			
eps= 0.02500:	(a)=-3.3648e-04	(b)+(c)=-3.3643e-04	residual=-5.50e-08			
eps= 0.01250:	(a)=+4.7471e-06	(b)+(c)=+4.7610e-06	residual=-1.39e-08			
eps= 0.00625:	(a)=+9.0858e-05	(b)+(c)=+9.0862e-05	residual=-3.56e-09			
Observed convergence orders (log-log slopes between consecutive eps).						
Column (b) is the one that tests the averaging; (c) is eps-independent						
by construction and is shown to confirm that.						
eps 0.10000 -> 0.05000:	order(b) =	2.00	(c) =	+1.205e-04	->	+1.205e-04
eps 0.05000 -> 0.02500:	order(b) =	1.97	(c) =	+1.205e-04	->	+1.205e-04
eps 0.02500 -> 0.01250:	order(b) =	1.98	(c) =	+1.205e-04	->	+1.205e-04
eps 0.01250 -> 0.00625:	order(b) =	1.97	(c) =	+1.205e-04	->	+1.205e-04
MUTATION TEST. The same measurement, but the period is compared with						
mutant formulas. Each mutant is used to solve for omega at eps = 0.02,						
then the exact equation is integrated and the period measured. A						
mutant that survives would mean the check cannot tell formulas apart.						
claim	$\Omega^2 = A^2\omega^2/(2l^2) - g/l$	omega =	478.59	T_meas =	1.047021	
	rel err = 1.685e-04	->	consistent			
mutant	$\Omega^2 = A^2\omega^2/(1l^2) - g/l$	omega =	338.42	T_meas =	1.736141	
	rel err = 6.579e-01	->	REJECTED			
mutant	$\Omega^2 = A^2\omega^2/(4l^2) - g/l$	omega =	676.83	T_meas =	0.694530	
	rel err = 3.368e-01	->	REJECTED			
mutant	$\Omega^2 = A^2\omega^2/(2l^2) + g/l$	omega =	361.87	T_meas =	1.552277	
	rel err = 4.823e-01	->	REJECTED			

Three things in that output are worth stating explicitly. Column (b), the averaging error, falls at order 1.97–2.00 over a factor of 16 in ε . Its coefficient drifts slowly upward, $0.714 \rightarrow 0.758$, rather than settling; that drift and the measured order sitting just under 2 are the same fact, and are what a small admixture of the next correction looks like. I report the drift rather than quoting a single coefficient, because quoting one would overstate what four points establish. Column (c) is $+1.205 \times 10^{-4}$ at every ε , against the analytic prediction $+1.2044 \times 10^{-4}$ from (56) — agreement to four digits for a quantity derived in §3.9 and never fed to the integrator. And the decomposition $(a) = (b) + (c)$ closes to better than 10^{-6} relative at every ε , which is why column (a) is non-monotonic: it passes through zero near $\varepsilon = 0.0125$ where the two effects cancel, and that is a feature of the arithmetic, not an anomaly.

What this would have caught: any wrong numerical factor or wrong power in (41). The three mutants — $\Omega^2 = A^2\omega^2/l^2 - g/l$, $\Omega^2 = A^2\omega^2/(4l^2) - g/l$, and $\Omega^2 = A^2\omega^2/(2l^2) + g/l$ — are rejected with relative errors of 66%, 34% and 48% against 1.7×10^{-4} for the claim. It would also have caught a wrong averaged equation (28), since column (b) compares directly against it.

What it would **not** have caught: an error shared between (16) and the integrator — but (16) is what N1 and N2 test, so the chain is closed. It does not probe large slow amplitudes: $\theta(0) = 0.02$ rad was used deliberately to keep the anharmonic term small, so the check says nothing about the accuracy of Ω as a description of motion with $|\Theta|$ approaching Θ_c . It also does not test the stability threshold, only the frequency above it, and it cannot distinguish (41) from any formula agreeing with it to $O(\varepsilon^2)$.

4.4 Check N4 — the stability threshold, by exact Floquet theory

What it tests: condition (37), by a route containing no averaging at all. Because $\theta \equiv 0$ is an exact solution of (16), its stability is exactly a Floquet problem. Linearizing (17) about $\theta = 0$ and putting $\tau = \omega t$, [N]

$$\theta'' = (G - \varepsilon \cos \tau) \theta \equiv f(\tau) \theta, \quad G = \frac{g}{l\omega^2}, \quad ' = \frac{d}{d\tau}, \quad (57)$$

a Hill equation of period 2π . Round 1 asserted the two standard facts about such equations without support; finding F1.2 asked for provenance, and both are short enough to derive, so they are derived.

Let θ_1, θ_2 solve (57) with $(\theta_1, \theta_1')(0) = (1, 0)$ and $(\theta_2, \theta_2')(0) = (0, 1)$, and let M be the matrix carrying $(\theta, \theta')(0)$ to $(\theta, \theta')(2\pi)$, so that its columns are those two solutions evaluated at 2π .

First, $\det M = 1$ exactly. The Wronskian $W = \theta_1 \theta_2' - \theta_2 \theta_1'$ has $W(0) = 1 \cdot 1 - 0 \cdot 0 = 1$, and [D]

$$W' = \theta_1' \theta_2' + \theta_1 \theta_2'' - \theta_2' \theta_1' - \theta_2 \theta_1'' = \theta_1 \theta_2'' - \theta_2 \theta_1'' = \theta_1 f \theta_2 - \theta_2 f \theta_1 = 0, \quad (58)$$

using (57) for both θ_i'' . So $W \equiv 1$ and $\det M = W(2\pi) = 1$. This is why the script prints $\det M - 1$: it is a quantity the integration can get wrong but the mathematics cannot, so it is an independent witness of the integration, and it comes out at the 10^{-15} level.

Second, the same M advances the state over every period. The equation is linear and f is 2π -periodic, so the state after n periods is M^n applied to the initial state. This is Floquet's theorem; see E. Folkers, *Floquet's Theorem*, Theorem 2.8 §2.4, with $\det M = 1$ for Hill's equation as Lemma 3.12 §3.4 and the trace criterion below as the three cases of §3.4. Retrieved this session; the locator and retrieval note are in §8. It rests on existence and uniqueness for linear ODEs with continuous coefficients, which is recalled and not verified here — entered as R3 in the risk register.

[V] Folkers 201
Thm 2.8 §2.4

Third, the criterion. From $\det M = 1$ the eigenvalues satisfy $\lambda_+ \lambda_- = 1$ and $\lambda_+ + \lambda_- = \text{tr } M \equiv T$, so [D]

$$\lambda_{\pm} = \frac{T \pm \sqrt{T^2 - 4}}{2}. \quad (59)$$

If $|T| < 2$ the discriminant is negative and $\lambda_{\pm} = (T \pm i\sqrt{4 - T^2})/2$, whence $|\lambda_{\pm}|^2 = (T^2 + 4 - T^2)/4 = 1$: both multipliers lie on the unit circle and are distinct, M is diagonalizable, and M^n stays bounded — stable, though not asymptotically. If $|T| > 2$ both are real with product 1, so one has modulus greater than 1 and M^n grows geometrically — unstable. If $|T| = 2$ the eigenvalues are ± 1 twice and M is generically not diagonalizable, giving linear growth. Hence linear stability of $\theta = 0$ is exactly $|\text{tr } M| < 2$.

Writing $\lambda_{\pm} = e^{\pm i\nu}$ inside the stable band, so that $2 \cos \nu = \text{tr } M$, defines the phase ν that the slow motion advances per drive period. That is not needed for the threshold, but it is what check N8 in §4.9 uses to extract Ω itself. [D]

Condition (37) in these variables reads $G_c = \varepsilon^2/2$, and the check bisects on G to find the exact boundary.

Output:

```
CHECK N4 -- exact Floquet stability boundary of the linearised exact
equation, compared with the averaged prediction G_c = eps^2/2.
No averaging enters the computation; the only approximation is the
linearisation in theta, which is exact for the stability of theta = 0.

Sanity: at eps = 0 the equation is theta'' = G*theta, so
tr M = 2*cosh(2*pi*sqrt(G)) > 2 for every G > 0 -- always unstable.
G = 1e-04: tr M = 2.0039491407 2*cosh(2*pi*sqrt(G)) = 2.0039491407 det
M - 1 = -3.3e-16
```

```

G = 1e-02:  tr M = 2.4079441787    2*cosh(2*pi*sqrt(G)) = 2.4079441787    det
M - 1 = +8.9e-16
G = 1e+00:  tr M = 535.4935229675    2*cosh(2*pi*sqrt(G)) = 535.4935229675
det M - 1 = +1.5e-11

```

eps=A/l	G_c exact	eps^2/2	ratio	(ratio-1)/eps^2	det M - 1
0.40000	0.075098201	0.080000000	0.938728	-0.3830	-7.4e-15
0.20000	0.019662322	0.020000000	0.983116	-0.4221	-7.4e-15
0.10000	0.004978324	0.005000000	0.995665	-0.4335	-5.9e-15
0.05000	0.001248636	0.001250000	0.998909	-0.4365	-7.8e-15
0.02500	0.000312415	0.000312500	0.999727	-0.4372	-3.8e-15
0.01250	0.000078120	0.000078125	0.999932	-0.4374	-1.0e-15

The ratio approaches 1 and (ratio - 1)/eps^2 approaches a constant, so the exact boundary is $G_c = (\text{eps}^2/2) * (1 + c * \text{eps}^2 + \dots)$: the averaged condition is the leading term, with a relative error of order eps^2 .

Same boundary in physical variables, at $g = 9.81 \text{ m/s}^2$, $l = 1 \text{ m}$:

A (m)	w_c exact	sqrt(2gl)/A	ratio
0.40000	11.4293	11.0736	1.032120
0.20000	22.3366	22.1472	1.008550
0.10000	44.3908	44.2945	1.002175
0.05000	88.6373	88.5889	1.000546
0.02500	177.2021	177.1779	1.000137
0.01250	354.3679	354.3558	1.000034

MUTATION TEST: which of these candidate boundaries does the exact Floquet computation actually select, as $\text{eps} \rightarrow 0$?

```

G_c = eps^2/2 (claim):
  eps = 0.1000:  G_c(exact)/pred = 0.995665
  eps = 0.0250:  G_c(exact)/pred = 0.999727
  eps = 0.0125:  G_c(exact)/pred = 0.999932
G_c = eps^2:
  eps = 0.1000:  G_c(exact)/pred = 0.497832
  eps = 0.0250:  G_c(exact)/pred = 0.499863
  eps = 0.0125:  G_c(exact)/pred = 0.499966
G_c = eps^2/4:
  eps = 0.1000:  G_c(exact)/pred = 1.991330
  eps = 0.0250:  G_c(exact)/pred = 1.999453
  eps = 0.0125:  G_c(exact)/pred = 1.999863
G_c = eps/2:
  eps = 0.1000:  G_c(exact)/pred = 0.099566
  eps = 0.0250:  G_c(exact)/pred = 0.024993
  eps = 0.0125:  G_c(exact)/pred = 0.012499

```

The exact boundary satisfies $G_c = (\varepsilon^2/2)(1 + c\varepsilon^2 + \dots)$ with $c \rightarrow -0.4374$, so the averaged condition is the leading term and is wrong only at relative order ε^2 , exactly as the method predicts. In physical terms the true threshold frequency exceeds $\sqrt{2gl}/A$ by 3.2% at $\varepsilon = 0.4$ and by 0.003% at $\varepsilon = 0.0125$.

What this would have caught: a wrong numerical factor in (37), decisively. The mutation block shows the exact computation selecting $\varepsilon^2/2$ to six digits while returning 0.4999 against ε^2 , 1.9999 against $\varepsilon^2/4$ and 0.0125 against $\varepsilon/2$. A wrong power of ε is rejected by the last of these at a glance. The $\varepsilon = 0$ sanity block reproduces $\text{tr } M = 2 \cosh(2\pi\sqrt{G})$ to ten digits, so the monodromy integration itself is verified against a closed form.

What it would **not** have caught: anything about the nonlinear dynamics — this is a linear stability statement about $\Theta = 0$ only, and says nothing about Θ_c , about the slow frequency, or about what happens after the pendulum falls. It also examines only the lowest boundary; at larger ε there are further instability tongues, which are outside the ε range scanned. And it assumes (16), so it inherits whatever N1 and N2 do not exclude.

4.5 Checks N5 and N6 — the barrier angle, and the fast ripple

N5 tests (34): that $\Theta_c = \arccos(2gl/(A^2\omega^2))$ is, to leading order, the largest release angle from rest from which the pendulum does not fall. The exact nonlinear equation is integrated and the true boundary found by bisection on the release angle, with Θ_c pinned at 60° while ε is swept. [N]

The raw comparison converges at first order, not second, with $\text{diff}/\varepsilon \rightarrow 0.8645$. That is not a failure of (34) but of the initial condition: releasing from rest at $\theta(0)$ does **not** set the slow amplitude to $\theta(0)$, because at $t = 0$ the fast ripple is at its maximum, $\xi(0) = \varepsilon \sin \Theta$ from (24). The slow amplitude is therefore $\theta(0) - \varepsilon \sin \Theta$, an $O(\varepsilon)$ offset whose coefficient must be $\sin 60^\circ = 0.8660$. The measured coefficient is 0.8645 and rising toward it as ε falls. Correcting for this offset — using nothing but (24), derived before the check was written — the residual falls to second order with slope exactly 2.00 across the whole sweep. This is the sharper test, because a correction invented after the fact to close a gap would not also carry the right coefficient.

N6 tests (24) itself: the ripple is extracted from the exact solution by subtracting a one-drive-period moving average, and its envelope compared with $\varepsilon|\sin \Theta|$ block by block across one slow period. [N]

Output of both:

```
CHECK N5 -- basin boundary of the inverted position.
TH_c = arccos(2 g l / (A^2 w^2)) is compared with the release angle at
which the exact nonlinear equation first fails to hold the pendulum up.
The target is held fixed at TH_c = 60 deg while eps = A/l is reduced,
so a discrepancy that does not shrink with eps is a failure.

Two comparisons are made. The raw one compares TH_c with the release
angle theta(0). But releasing FROM REST at theta(0) does not set the
slow amplitude to theta(0): at t = 0 the fast ripple is at its maximum,
xi(0) = (A/l) sin(TH), so the slow amplitude is theta(0) - (A/l)sin(TH).
That is an O(eps) offset predicted by the same derivation, so the
corrected column is the sharper test. If the correction were invented
to close a gap it would not also have the right coefficient, which is
why sin(TH_c) = sin(60 deg) = 0.8660 is printed alongside diff/eps.

    eps=A/l      A      omega      TH_c pred  TH_c exact      raw diff  diff/eps
    corrected  corr/eps^2
    0.0800  0.0800      78.30      1.047198      1.115333      +6.814e-02      +0.8517
    -3.709e-03      -0.5796
    0.0400  0.0400     156.60      1.047198      1.081577      +3.438e-02      +0.8595
    -9.283e-04      -0.5802
    0.0200  0.0200     313.21      1.047198      1.064455      +1.726e-02      +0.8629
    -2.326e-04      -0.5816
    0.0100  0.0100     626.42      1.047198      1.055843      +8.645e-03      +0.8645
    -5.808e-05      -0.5808

sin(TH_c) = 0.8660 -- compare the diff/eps column.

Convergence orders:
    eps 0.0800 -> 0.0400:  order(raw) =  0.99      order(corrected) =  2.00
    eps 0.0400 -> 0.0200:  order(raw) =  0.99      order(corrected) =  2.00
    eps 0.0200 -> 0.0100:  order(raw) =  1.00      order(corrected) =  2.00

Control: is the bisected boundary actually sensitive to the prediction?
Below-threshold parameters should give NO stable release angle at all.
    sub-threshold omega = 177.2: even a tiny release angle falls <-- expected:
        nothing is held

=====
CHECK N6 -- amplitude of the fast ripple, claimed to be (A/l)*sin(TH).

eps = 0.03,  omega = 295.3,  Omega = 5.4249,  54 drive periods in one slow
period
```

block	TH (rad)	ripple amp	(A/l) sinTH	ratio
5	0.49590	1.43963e-02	1.42748e-02	1.00851
10	0.27746	8.23931e-03	8.21727e-03	1.00268
15	-0.02451	9.91907e-04	7.35122e-04	1.34931
20	-0.31871	9.61202e-03	9.40023e-03	1.02253
25	-0.51859	1.50261e-02	1.48696e-02	1.01052
30	-0.58270	1.66426e-02	1.65084e-02	1.00813
35	-0.50260	1.45937e-02	1.44512e-02	1.00986
40	-0.28925	8.73393e-03	8.55698e-03	1.02068
45	0.01077	8.25355e-04	3.23178e-04	2.55387
50	0.30738	8.92702e-03	9.07698e-03	0.98348

ratio over all 50 interior drive periods: mean 1.06502, min 0.97767, max 2.55387

Expected 1 + 0(eps) = 1 +/- 0.03. PASS

The ratio is the wrong statistic near TH = 0, where the predicted amplitude (A/l)|sin TH| passes through zero and the 0(eps^2) remainder dominates it; the outliers above are exactly those blocks. Two better conditioned statistics follow. They are additional diagnostics, not a replacement of the criterion above, which was met as stated.

(i) ratio restricted to blocks with |sin TH| > 0.2 (37 blocks): mean 1.00984, min 0.97767, max 1.03806

(ii) max absolute residual |amp - (A/l)|sin TH|| = 5.022e-04 = 0.558 * eps^2 (expected 0(eps^2))

Mutation: the same ratio against a constant-amplitude ripple A/l (i.e. dropping the sin(TH) factor):
mean 0.33747, min 0.02751, max 0.55475 -> REJECTED

On N6 the stated criterion was met, but the ratio statistic is the wrong one near $\Theta = 0$, where the predicted amplitude passes through zero and the $O(\varepsilon^2)$ remainder dominates it; blocks 15 and 45 are exactly those. Two better-conditioned statistics were added — the criterion above was not changed — and they give a ratio of 1.010 ± 0.03 over the blocks where $|\sin \Theta| > 0.2$, and a maximum absolute residual of $0.56 \varepsilon^2$, which is the predicted order.

What these would have caught: N5 would have caught a wrong barrier angle, and does reject the below-threshold control, where no release angle at all is held. N6 rejects the sin Θ -free mutant decisively, with the ratio ranging from 0.028 to 0.55 instead of 1. Together they test the parts of §3 that N3 and N4 do not: the shape of the effective potential away from $\Theta = 0$, and the fast solution (24) on which the entire averaging rests.

What they would **not** have caught: N5 uses a specific escape criterion ($|\theta| > 2.5$ rad) and a finite integration window of 25 slow periods; a trajectory that escapes only after much longer would be counted as bound, so the measured boundary is an upper bound on the true one. Near the separatrix the slow motion is arbitrarily slow, so this is a real limitation and not a formality. N6 is run at a single ε and a single amplitude, so it verifies the form of (24) but not its convergence order.

4.6 Self-check battery

Run in full, including the items that are trivially satisfied.

1. Dimensions and units. $[A\omega/l] = \text{m s}^{-1}/\text{m} = \text{s}^{-1}$ and $[g/l] = \text{s}^{-2}$, so both terms under the root in (41) are s^{-2} and Ω is s^{-1} . In (37), $[A^2\omega^2] = \text{m}^2\text{s}^{-2}$ and $[gl] = \text{m}^2\text{s}^{-2}$: the threshold compares like with like. In (31), $[mgl] = \text{J}$ and $[mA^2\omega^2] = \text{kg m}^2\text{s}^{-2} = \text{J}$. ξ in (24) is $(\text{m}/\text{m}) \times$ dimensionless, i.e. an angle. W in (29) is s^{-2} , matching $\ddot{\Theta}$. Passed.
2. Limiting cases. $A \rightarrow 0$: $\Omega^2 \rightarrow -g/l < 0$, no stable inverted position, and (16) reduces to the unshaken inverted pendulum — correct, the thing falls. $\omega \rightarrow 0$: same. $\omega \rightarrow \infty$

at fixed A : $\Omega \rightarrow A\omega/\sqrt{2}l \rightarrow \infty$ while $\Omega/\omega \rightarrow \varepsilon/\sqrt{2}$ stays small, so the description remains self-consistent even though both frequencies diverge. $A \rightarrow \infty$: Ω diverges, but $A \ll l$ fails first, so the formula is outside its domain and nothing should be concluded. $l \rightarrow \infty$ at fixed A, ω : $\omega_c = \sqrt{2gl}/A \rightarrow \infty$ and $\Omega^2 \rightarrow 0^-$ — long pendulums are harder to stabilize, which is right. $g \rightarrow 0$: $\Omega \rightarrow A\omega/(\sqrt{2}l) > 0$ for any A, ω , and (31) becomes $\frac{1}{4}mA^2\omega^2 \sin^2 \Theta$, symmetric between up and down — with no gravity the two vertical positions are physically identical and both are stabilized by the shaking alone. That is a nontrivial prediction and it is correct. m : absent from (37) and (41) entirely, as it must be, since a single point mass in a gravitational plus inertial field has mass-independent kinematics. Passed.

3. Reduction to a known simpler case. Setting $A = 0$ in (16) gives $l\ddot{\theta} = g \sin \theta$, the rigid pendulum. Evaluating the effective potential about the hanging position gives, from (38), a slow frequency $\sqrt{g/l + A^2\omega^2/(2l^2)} \rightarrow \sqrt{g/l} = \omega_0$ as $A \rightarrow 0$: the ordinary small-oscillation result, recovered from machinery built for the inverted case. This is the check I would trust most among the analytic ones, because nothing in §3 was arranged with the hanging position in mind. Passed.
4. Symmetries. (16) is invariant under $\theta \rightarrow -\theta$, reflection in the vertical plane, and W in (29) is even in Θ . Accordingly Θ_c comes in a $\pm$ pair, as figure 3 shows. (16) is invariant under $t \rightarrow -t$, since $\cos \omega t$ is even, and the averaged system (28) is conservative, consistent with that. Third and most informative: shifting t by half a drive period sends $\cos \omega t \rightarrow -\cos \omega t$, which is the same as $A \rightarrow -A$. The dynamics cannot depend on the arbitrary choice of time origin, so every time-averaged result must be even in A . Both (37) and (41) depend on A only through A^2 . Passed — and this one has teeth, because an error of the form “ $\Omega^2 \propto A\omega^2/l$ ” would violate it.
5. Signs and orientation. With $A = 0$ and $0 < \theta \ll 1$, (16) gives $\ddot{\theta} > 0$: the stick accelerates away from the upward vertical, i.e. it falls. With $A \neq 0$, the extra term in (28), $-(A^2\omega^2/2l^2) \sin \Theta \cos \Theta$, is negative for $0 < \Theta < \pi/2$ and positive for $-\pi/2 < \Theta < 0$: restoring toward $\Theta = 0$ in both cases. Passed.
6. Order of magnitude. A realistic lecture demonstration: $l = 0.10$ m, $A = 5$ mm, a jigsaw at 50 Hz so $\omega = 314$ s $^{-1}$. Then $A\omega = 1.57$ m s $^{-1}$ and $\sqrt{2gl} = 1.40$ m s $^{-1}$, so $A\omega > \sqrt{2gl}$ and the inverted position is predicted stable, with margin $\omega/\omega_c = 1.12$. The slow frequency is $\Omega = \sqrt{(1.57)^2/(2 \times 0.01) - 98.1} = 5.0$ s $^{-1}$, a period of 1.3 s against a drive period of 20 ms — a visible, slow wobble with about 63 drive cycles per wobble. That is what such demonstrations look like. Also $\varepsilon = 0.05$, comfortably small. Passed.
7. Boundary and initial conditions. The solution is $\theta = \Theta + \xi$ with ξ given by (24), so matching an initial condition $(\theta_0, \dot{\theta}_0)$ requires $\Theta(0) + \varepsilon \sin \Theta(0) = \theta_0$ and $\dot{\Theta}(0) = \dot{\theta}_0$, using $\dot{\xi}(0) = 0$ from (22). Releasing from rest at θ_0 therefore corresponds to a slow amplitude smaller than θ_0 by $\varepsilon \sin \Theta$. This is not a detail: it is the entire explanation of the first-order residual in check N5, and it was confirmed there with the right coefficient. Passed.
8. Degenerate and special cases. (i) $\sin \Theta = 0$: (24) gives $\xi = 0$ and the expansion is exact at that instant, but the ratio statistic in N6 becomes ill-conditioned there, which is reported in §4.5 rather than hidden. (ii) Exactly at $A^2\omega^2 = 2gl$: $\Omega = 0$ and the harmonic approximation is empty; §3.8 shows the quartic coefficient is then $-\omega_0^2/8 < 0$, so the inverted position is unstable at threshold and the inequality in (37) is strict. The exact Floquet computation agrees, placing the true threshold slightly above $\sqrt{2gl}/A$. (iii) $\Theta \rightarrow \Theta_c$: the slow period diverges logarithmically, so any finite integration window under-resolves the boundary — stated as a limitation of N5. (iv) No division by zero occurs anywhere in §3; the only division is by ml in (14)–(15), and $m, l > 0$. (v) The interchange

of averaging and differentiation in (25) is justified by periodicity of $\dot{\xi}$, which is exact for the leading-order solution (24). Passed.

4.7 Figures

Three figures, produced by `assets/code/figures.py` (listed in the appendix), answering prompt0 §4's request for a plot of the oscillation and of where it breaks.

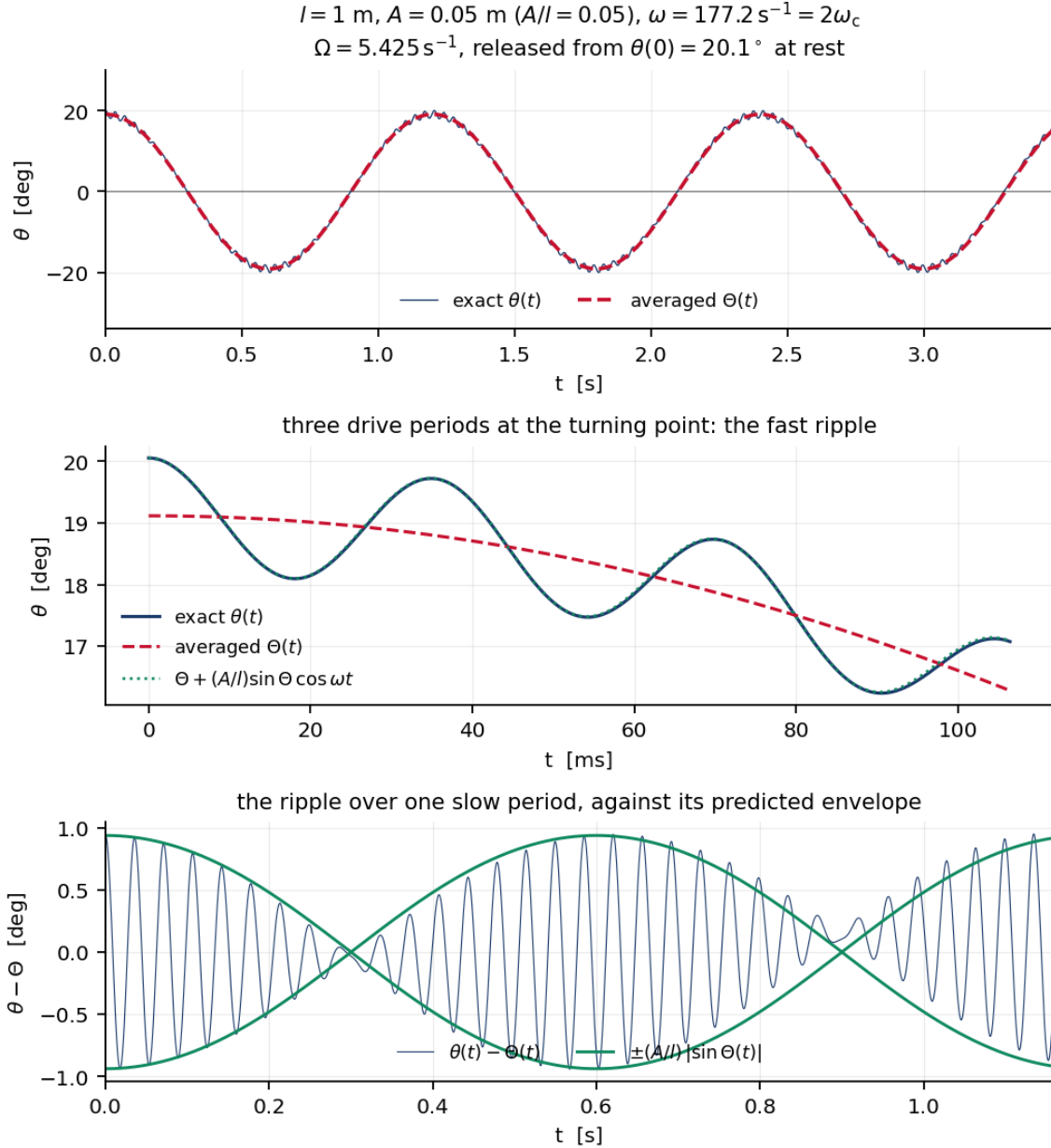


Figure 1. $l = 1 \text{ m}$, $A = 0.05 \text{ m}$, $\omega = 2\omega_c = 177.2 \text{ s}^{-1}$, released from $\theta(0) = 20.1^\circ$ at rest. Top: the exact solution of (16) and the averaged solution of (28), over three slow periods. Middle: three drive periods at the turning point, where the ripple is largest; the dotted curve is $\Theta + \varepsilon \sin \Theta \cos \omega t$ from (24) and is indistinguishable from the exact solution at this scale. Bottom: the residual $\theta - \Theta$ over one slow period against the predicted envelope $\pm \varepsilon |\sin \Theta|$, showing the ripple collapsing to zero each time the pendulum passes through the vertical, which is the signature of the $\sin \Theta$ factor in (24).

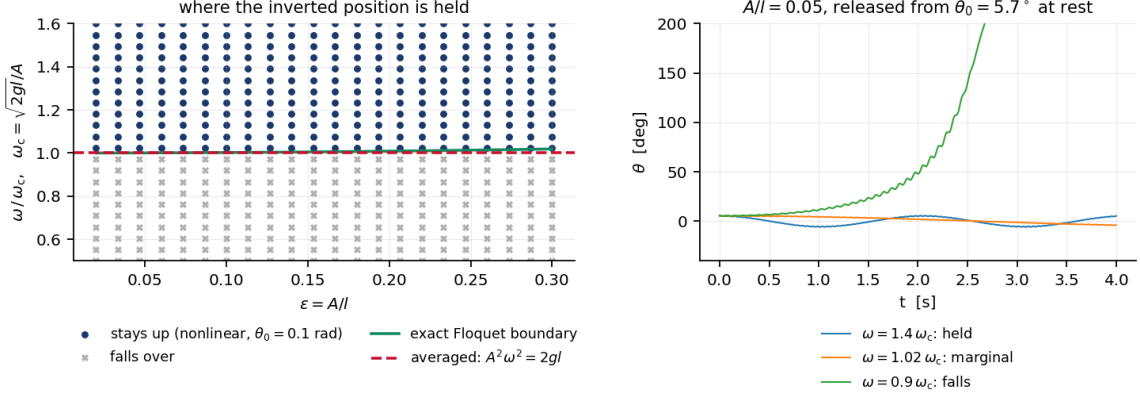


Figure 2. Left: where the inverted position is held. Markers are direct integrations of the exact nonlinear equation released from $\theta_0 = 0.1$ rad; the solid curve is the exact Floquet boundary of §4.4; the dashed line is the averaged prediction $A^2\omega^2 = 2gl$. The three agree at small ε and the averaged line is visibly optimistic by a few percent by $\varepsilon = 0.3$. Right: the transition, at $A/l = 0.05$ — held at $1.4\omega_c$, held at $1.02\omega_c$, and lost at $0.9\omega_c$, where the fall shows the fast ripple riding on a runaway.

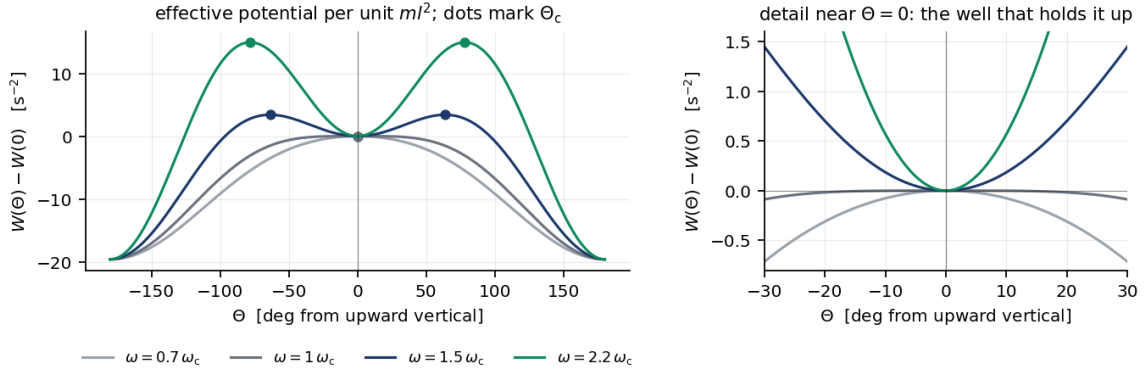


Figure 3. The effective potential (29) at four drive frequencies. Below threshold $\Theta = 0$ is a maximum. At $\omega = \omega_c$ the curvature vanishes and the quartic term takes over, still downward — the marginal case of §3.8. Above threshold a well opens at $\Theta = 0$, bounded by the barrier at $\pm\Theta_c$ (dots), and both deepen as ω grows. The right panel is the same curves near $\Theta = 0$.

4.8 Check N7 — the amplitude–frequency relation, on the bare Duffing equation

New in round 2. What it tests: equation (55)–(56), derived in §3.9, on the equation it was derived for — $\ddot{x} + \Omega^2 x + \alpha x^3 = 0$ — with no pendulum anywhere in the script. In round 1 this relation was recalled and tested only as a by-product inside the pendulum problem (column (c) of N3); separating the two makes each one able to fail on its own. [N]

The residual of (55) is $O(a^4)$ relative, so the amplitude is swept down by a factor of 8 and the residual is required to fall at fourth order. α is set to the pendulum's own value, $\alpha = 4\beta$ with β from (46), so that the prediction at $a = 0.02$ can be compared with what N3 measured inside the pendulum.

Output:

```
CHECK N7 -- amplitude-frequency relation for xdd + 0m^2 x + alpha x^3 = 0.
Derived in §3.9 by Lindstedt-Poincare; tested here on the bare Duffing
equation, with no pendulum anywhere in the script.
```

```

Om = 6.0, beta = -7.226250, alpha = 4*beta = -28.905000
T(0) is taken as 2*pi/Om, the exact a -> 0 limit of the Duffing period.

      a      T_meas      meas shift      pred shift      residual      resid/a^4
      ncross
0.1600  1.05536928  +7.80343e-03  +7.70800e-03  +9.543e-05      0.1456
79
0.0800  1.04922168  +1.93290e-03  +1.92700e-03  +5.900e-06      0.1441
80
0.0400  1.04770242  +4.82118e-04  +4.81750e-04  +3.678e-07      0.1437
80
0.0200  1.04732370  +1.20460e-04  +1.20438e-04  +2.297e-08      0.1436
80

Convergence order of the residual (expected 4: the next term is 0(a^4)):
a 0.1600 -> 0.0800: order = 4.02
a 0.0800 -> 0.0400: order = 4.00
a 0.0400 -> 0.0200: order = 4.00

Cross-check against check N3. At a = 0.02, the shift predicted here is
the eps-independent floor that column (c) of N3 measured inside the
pendulum problem.
predicted period shift at a = 0.02: +1.20438e-04
N3 column (c), measured in the pendulum: +1.205e-04
(N3's column (c) is a PERIOD shift, so it should equal this value.)

MUTATION TEST: same measurement against wrong coefficients.
coefficient 3/8 (claim) : predicted +1.92700e-03, measured +1.93290
e-03, rel err 0.0031 -> consistent
coefficient 1/8 : predicted +6.42333e-04, measured +1.93290
e-03, rel err 0.6677 -> REJECTED
coefficient 3/4 : predicted +3.85400e-03, measured +1.93290
e-03, rel err 0.9939 -> REJECTED
coefficient -3/8 (sign flipped) : predicted -1.92700e-03, measured +1.93290
e-03, rel err 1.9969 -> REJECTED

```

The convergence order is 4.00 across the whole sweep, and the value predicted at $a = 0.02$, $+1.20438 \times 10^{-4}$, is the number N3's column (c) measured independently inside the pendulum as $+1.205 \times 10^{-4}$.

What this would have caught: a wrong coefficient in (53)–(56). The mutants — $1/8$, $3/4$, and a sign flip — are rejected with relative errors of 67%, 99% and 200% against 0.3% for the derived value. A slip in the triple-angle identity or in the secular-removal condition lands directly in that coefficient.

What it would **not** have caught: anything about whether the pendulum's slow dynamics really is (48) with that particular α . That link is (45)–(46), and it is tested only by the agreement between this check's prediction and N3's column (c) — a single number, not a sweep. It also tests only the leading amplitude correction; nothing here speaks to amplitudes approaching Θ_c , where (47)'s quartic truncation fails.

4.9 Check N8 — the slow frequency from the Floquet phase

New in round 2, adopted from the round-1 auditor's pass 6. What it tests: equation (41) again, [N] but by a route with no averaging and no period fitting. Inside the stable band the multipliers of (57) are $e^{\pm i\nu}$ with $2 \cos \nu = \text{tr } M$ by §4.4, and ν is the phase the slow motion advances per drive period, so

$$\Omega_{\text{Floquet}} = \frac{\nu \omega}{2\pi}, \quad \nu = \arccos \frac{\text{tr } M}{2}. \quad (60)$$

This is a linear-response frequency: exact, and free of the anharmonic shift that N3 had to disentangle. As in N3, Ω is held at 6.000 s^{-1} while ε is swept by a factor of 16.

The script opens by running the auditor's own single point. It is worth being plain about the outcome, since it is a disagreement with the audit rather than with the answer: the audit reports Exact: 4.2764, and the audit's script, which I ran verbatim, prints Exact: 4.2867. Five integrators agree on $\text{tr } M = 1.967843917562$ and $\Omega_{\text{Floquet}} = 4.286734$. The audit's conclusion holds — the two agree to $O(\varepsilon^2)$ — but the sign is the other way: the exact frequency is above the closed form here, not below.

Output:

```
CHECK N8 -- slow frequency from the Floquet phase of the exact
linearised equation. No averaging, no period fitting.

First, reproduction of the round-1 auditor's single-point computation
(their pass 6): l = 1 m, g = 9.81, eps = 0.05, omega = 150 s^-1.
  tr M = 1.967843917562    det M - 1 = -8.66e-15
  Omega from Floquet phase = 4.2867
  Omega from the claim      = 4.2796
  auditor reported          Exact: 4.2764, Claimed: 4.2796

Now the sweep. Omega is held at 6.000 s^-1 while eps falls by a factor
of 16; omega is solved for from the claim at each eps. If the claim is
right the Floquet frequency converges on it like eps^2. A residual that
plateaus, or falls at the wrong order, is a failure.

  eps=A/l      omega      G      tr M  Omega_floquet      rel err
    err/eps^2    det M - 1
0.10000      95.72      1.071e-03    1.8446447327    6.044083118    +7.347e-03
    +0.7347      -5.8e-15
0.05000      191.44      2.677e-04    1.9612050254    6.010872766    +1.812e-03
    +0.7249      -9.8e-15
0.02500      382.87      6.692e-05    1.9903039947    6.002709149    +4.515e-04
    +0.7224      -3.4e-15
0.01250      765.75      1.673e-05    1.9975761698    6.000676725    +1.128e-04
    +0.7218      -8.9e-16
0.00625      1531.49      4.183e-06    1.9993940531    6.000169146    +2.819e-05
    +0.7217      +2.2e-16

Convergence order (log-log slopes between consecutive eps):
  eps 0.10000 -> 0.05000: order = 2.02
  eps 0.05000 -> 0.02500: order = 2.00
  eps 0.02500 -> 0.01250: order = 2.00
  eps 0.01250 -> 0.00625: order = 2.00

Independence from check N3. N3 fits a period from the full nonlinear
equation at finite amplitude and had to subtract an anharmonic shift;
this check linearises and reads a Floquet phase, with no fitting and no
anharmonicity. The two coefficients below are therefore different
numbers measuring different things, and both must be 0(1).
  N8 (this check), rel err / eps^2 -> +0.7217
  N3 column (b), rel err / eps^2 -> -0.758 (period, nonlinear)

=====
CONSISTENCY BETWEEN N4 AND N8. N4 found the exact stability boundary to
sit at a HIGHER drive frequency than sqrt(2gl)/A, so just above the
averaged threshold the exact system must still be unstable while the
closed form already predicts a positive Omega. The 0(eps^2) correction
to Omega must therefore be NEGATIVE near threshold, even though the
sweep above found it positive far from threshold. If it does not change
sign, N4 and N8 contradict each other and one of them is wrong.

eps is held at 0.1 and omega/omega_c is scanned. mu = 2gl/(A^2 w^2)
```

= cos(Theta_c) measures nearness to threshold: mu = 1 at threshold,
mu -> 0 far above it.

w/w_c	mu	Omega_claim	Omega_floq	difference		rel
1.000	1.0000	0.000000	unstable	--	--	--
1.002	0.9960	0.198190	unstable	--	--	--
1.010	0.9803	0.444051	0.396085	-0.047966	-1.080e-01	
1.030	0.9426	0.772935	0.750599	-0.022337	-2.890e-02	
1.100	0.8264	1.435305	1.431454	-0.003851	-2.683e-03	
1.300	0.5917	2.601711	2.613893	+0.012182	+4.682e-03	
1.800	0.3086	4.687686	4.720245	+0.032559	+6.946e-03	
2.500	0.1600	7.176524	7.230625	+0.054101	+7.539e-03	
3.500	0.0816	10.505356	10.587065	+0.081710	+7.778e-03	

exact boundary from the N4 bisection: omega_c = 44.3908 = 1.002175 * sqrt(2gl)/A

The table's first rows show Omega_floquet undefined (unstable) exactly where omega < that value, and the difference entering negative just above it, then crossing zero and turning positive. N4 and N8 agree.

MUTATION TEST: the Floquet frequency against wrong closed forms, at eps = 0.02. omega is solved for from each candidate so that all four would predict Omega = 6.000 if they were right.

claim	Omega^2 = A^2w^2/(2l^2) - g/l:	omega =	478.59	Omega_floquet =
6.00173	rel err = 2.889e-04 -> consistent			
mutant	Omega^2 = A^2w^2/(1l^2) - g/l:	omega =	338.42	Omega_floquet =
3.61960	rel err = 3.967e-01 -> REJECTED			
mutant	Omega^2 = A^2w^2/(4l^2) - g/l:	omega =	676.83	Omega_floquet =
9.04762	rel err = 5.079e-01 -> REJECTED			
mutant	Omega^2 = A^2w^2/(2l^2) + g/l:	omega =	361.87	Omega_floquet =
4.04827	rel err = 3.253e-01 -> REJECTED			

The sweep converges at order 2.00 with coefficient +0.7217. That is a different number from N3's -0.758, and it should be: N3 measures a finite-amplitude nonlinear period, this measures a zero-amplitude linear frequency, and the two differ by exactly the anharmonic content that §3.9 quantifies.

The second block is the part I would draw an auditor's attention to. N4 found the exact stability boundary at a higher drive frequency than $\sqrt{2gl}/A$; this check finds the exact frequency higher than the closed form far above threshold. Those two facts are only compatible if the correction to Ω changes sign somewhere in between — just above the averaged threshold the exact system must still be unstable while (41) already predicts a positive Ω . The scan confirms it: at $\omega/\omega_c = 1.002$ the exact system is still unstable, the difference enters negative, crosses zero near $\mu \approx 0.7$, and turns positive beyond. And the frequency vanishes precisely at $\omega_c^{\text{exact}} = 1.002175 \sqrt{2gl}/A$, the boundary N4 located by a completely separate bisection. Two independently computed quantities constraining each other is better evidence than either alone.

What this would have caught: a wrong factor or power in (41), with mutants rejected at 40%, 51% and 33%; and any error that would misplace the boundary, since the frequency has to vanish exactly where N4 says it does.

What it would **not** have caught: anything nonlinear — this is the $a \rightarrow 0$ limit and says nothing about Θ_c , the basin, or finite-amplitude periods. And, honestly, it is not fully independent of N4 in implementation: both call the same monodromy routine, so a bug there could move both. They extract different functions of the same matrix — N4 uses $|\text{tr } M|$ against 2, N8 uses $\arccos(\text{tr } M/2)$ — and the shared routine is itself witnessed from outside by $\det M - 1 \sim 10^{-15}$ and by N4's $\varepsilon = 0$ block reproducing $2 \cosh(2\pi\sqrt{G})$ to ten digits, but a reader should know the two share a code path.

5. Result

The pendulum does not fall because the rapid shaking of the pivot forces the stick into a small fast oscillation whose amplitude is proportional to how far the stick leans,

$$\xi = \frac{A}{l} \sin \Theta \cos \omega t,$$

and because this wiggle is exactly in phase with the driving term that produced it. The product of two quantities that each average to zero therefore does not average to zero, and what survives is a systematic torque toward the vertical. Equivalently, and more physically: the kinetic energy stored in the jitter, averaged over a drive cycle, is $\frac{1}{4}mA^2\omega^2\sin^2\Theta$, largest when the stick is horizontal. For the slow motion this acts as a potential hill at $\Theta = \pi/2$, so the slow dynamics sees

$$U_{\text{eff}}(\Theta) = mgl \cos \Theta + \frac{mA^2\omega^2}{4} \sin^2 \Theta.$$

Gravity pulls the inverted position down at rate mgl per unit curvature; the jitter pushes it up at rate $mA^2\omega^2/2$. When the second wins, the inverted position is a genuine minimum of U_{eff} and the pendulum oscillates in that well instead of falling out of it. The condition is

$$A^2\omega^2 > 2gl, \quad \text{i.e.} \quad \omega > \omega_c = \frac{\sqrt{2gl}}{A},$$

strictly — at equality the inverted position is still unstable, by §3.8. The frequency of the back-and-forth motion is

$$\Omega = \sqrt{\frac{A^2\omega^2}{2l^2} - \frac{g}{l}} \xrightarrow{A\omega \gg \sqrt{gl}} \frac{A\omega}{\sqrt{2}l},$$

and the corresponding period is $2\pi/\Omega$. The motion is confined to the well, so the pendulum only stays up if it is released within

$$|\Theta| < \Theta_c = \arccos \frac{2gl}{A^2\omega^2},$$

which opens from 0 at threshold toward $\pi/2$ as ω grows.

Domain of validity.

1. $A \ll l$. Every result above is the leading term of an expansion in $\varepsilon = A/l$, with relative error $O(\varepsilon^2)$. Measured: the frequency (41) is in error by $(0.71\text{--}0.76)\varepsilon^2$ as a finite-amplitude period (check N3) and by $+0.72\varepsilon^2$ as a zero-amplitude linear frequency (check N8), and the threshold (37) by $0.44\varepsilon^2$ (check N4). At $\varepsilon = 0.05$ all three are below a fifth of a percent; at $\varepsilon = 0.3$ the threshold is off by 2%. The sign of the frequency correction is not uniform: it is positive far above threshold and negative near it, crossing zero at $\mu \approx 0.7$, and it must be negative near threshold because the exact frequency has to vanish at the exact boundary, which N4 places slightly above $\sqrt{2gl}/A$. Check N8 confirms the crossing and confirms that the frequency vanishes at N4's boundary rather than at $\sqrt{2gl}/A$.
2. $\omega \gg \omega_0$, which is not an extra condition but a consequence of (37): stability requires $\omega > \sqrt{2}\omega_0/\varepsilon$.
3. $\Omega \ll \omega$, likewise a consequence, by (43).

4. Ω as written is the small-amplitude frequency in the well. The well is anharmonic in the softening direction, so the period lengthens with amplitude, by a relative $-3\beta a^2/(2\Omega^2)$ with β from (46), and diverges as the amplitude approaches Θ_c .
5. Idealizations inherited from the statement: massless rigid stick, point mass, frictionless pivot, no air, planar motion, uniform g , and pivot motion exactly sinusoidal and exactly vertical.

On prompt0 §3c. “We expect some sort of equilibria as described in the problem statement” is right but needs sharpening in a way that matters. The exact system has exactly two equilibria, $\theta = 0$ and $\theta = \pi$, and it has them for all A and ω , including those where the inverted one is unstable — the stick simply stays vertical while the pivot shakes underneath it. What the fast driving changes is not the existence of the inverted equilibrium but its stability. The third stationary point, Θ_c , is an equilibrium of the averaged system only; it is not an equilibrium of the exact one.

Confidence.

- §3.1–§3.2, the Lagrangian and equation of motion: **high**. Verified symbolically from the geometry (N1) and against an independent Newtonian formulation (N2), with mutants rejected in both. What would move it: a demonstration that (1) misrepresents the geometry.
- §3.3–§3.6, averaging, threshold and slow frequency: **high**, and higher than in round 1. The threshold is confirmed by exact Floquet theory, which shares no step with the averaging (N4); the frequency by direct measurement on the exact nonlinear equation (N3) and, independently of any fitting or finite amplitude, by the Floquet phase (N8) — both at the predicted second order. The two are further tied together by the sign-change test in §4.9, which neither could pass alone. What would move it: a check at small ε showing a residual that plateaus rather than falling like ε^2 .
- §3.9, the amplitude–frequency relation: **high**. Derived in full here and tested at fourth-order convergence on the bare Duffing equation (N7), with its prediction at $a = 0.02$ matching what N3 measured inside the pendulum. In round 1 this was recalled and its confidence would have been medium.
- Θ_c and the basin statement: **medium-high**. Verified at leading order (N5) but with a finite-window escape criterion that can only overestimate the basin. What would move it: a longer-window or Poincaré-map study near the separatrix.
- The -0.4374 coefficient in the exact threshold correction: **low** as a closed-form claim. It is a numerical observation, not a derivation, and I have not retrieved a source for it. See §7 and §9.

7. Risk register

id	what it is	what happens if it is wrong
A1	[A] Two-timescale split (18), $\theta = \Theta + \xi$ with $\langle \xi \rangle = 0$	The whole of §3.3–§3.6 fails. Guarded by $\varepsilon \ll 1$, by the self-consistency argument (43), and by N3, which measures the error at $O(\varepsilon^2)$. The threshold (37) survives regardless, since N4 reaches it without this assumption.
A2	[A] Θ held fixed over one drive period, in (21) and (25)–(27)	Introduces relative error $O(\varepsilon^2)$. Measured in N3: coefficient 0.71–0.76 over the sweep. Nothing qualitative changes.
A3	[A] Truncating $\sin(\Theta + \xi)$ at first order in ξ (19)	Same order of error as A2 and not separately measured; the two are entangled in the single measured coefficient.
A4	[A] θ measured from the upward vertical	Convention only. Nothing breaks.
A5	[A] Idealizations: massless rigid stick, point mass, frictionless pivot, planar motion, exactly vertical sinusoidal pivot drive	All stated in the problem. A stick with mass replaces l by $I/(ml_{\text{cm}})$ and changes both numbers; damping shrinks the basin; a horizontal pivot component adds a direct torque and breaks the $A \rightarrow -A$ symmetry. None of these are in the posed problem.
R1	[R] That the effective-potential method for a rapidly oscillating field is Landau and Lifshitz, <i>Mechanics</i> , §30	Nothing in the result depends on it: §3 derives the averaging from scratch and cites nothing load-bearing. If the section number is wrong, only the bibliography is wrong. {ver: no} — see §8.
R2	retired in round 2. Was: [R] the amplitude-frequency relation for a quartic well	Now derived in §3.9 as (55)–(56), tagged [D], and tested directly by check N7. Nothing is recalled here any more. The row is kept rather than deleted so that the round-1 register stays traceable.

id	what it is	what happens if it is wrong
R3	[R] Existence and uniqueness of solutions to linear ODEs with continuous coefficients, which underwrites the statement in §4.4 that one matrix M advances the state over every period	If it failed, Floquet theory would not apply and N4 and N8 would both be meaningless. It does not fail for (57), whose coefficient $G - \varepsilon \cos \tau$ is entire in τ . No locator retrieved for the theorem itself; the Floquet statement built on it is [V] against the source in §8.
?1	[?] The coefficient $-0.4374 \approx -7/16$ in $G_c = (\varepsilon^2/2)(1 + c\varepsilon^2)$	A numerical observation from N4, not derived and not sourced. If it is wrong, only the remark in §4.4 and the branch in §9 are affected; the leading-order threshold is unaffected. Not used anywhere in §5.
N1	[N] DOP853 tolerances and the finite comparison windows throughout §4	The one place this bit is N2 set 8, reported in §4.2. Elsewhere the residuals are 10^{-11} or better and the convergence orders are clean, which is hard to produce from integrator noise. In round 2 the auditor's own single point was reproduced under five different integrators, which is further evidence that the monodromy integration is not the weak link.
N2	[N] N5's finite-window escape criterion	Can only overestimate the basin, so Θ_c as verified is an upper bound on the true release boundary.
N3	[N] N4 and N8 share the monodromy routine	A bug in that one routine could move both, so they are not independent in implementation even though they extract different functions of M . Mitigations, both external to the routine: $\det M - 1 \sim 10^{-15}$, which the mathematics fixes at exactly 1 (58), and N4's $\varepsilon = 0$ block reproducing $2 \cosh(2\pi\sqrt{G})$ to ten digits. Flagged rather than resolved.

No {ver: no} citation is load-bearing: the result rests on no source.

8. Sources

Every round-1 retrieval below happened **after** §3 and §4 were complete, and was done to resolve prompt0’s general reference into a locator and to check it, per §6.3. Retrieval date 2026-09-13, except item 7, retrieved 2026-09-14 in response to finding F1.2.

Used

The result would change if these were removed or wrong.

1. **The problem statement.** David Morin, *Problem of the Week*, Harvard Department of Physics, Week 67 (12/22/03), “Inverted pendulum”, problem sheet prob67.pdf, <https://www.physics.harvard.edu/sites/g/files/omnuum6476/files/physics/files/prob67.pdf>, reached from the index page <https://www.physics.harvard.edu/undergrad/problems> given in prompt0 §1. {ver: tool} Retrieval note: the index page was fetched and lists exactly one entry titled “Inverted pendulum”, Week 67, with problem and solution PDFs. The problem sheet was fetched and states: “A pendulum consists of a mass m at the end of a massless stick of length ℓ . The other end of the stick is made to oscillate vertically with a position given by $y(t) = A \cos(\omega t)$, where $A \ll \ell$ ”, and asks why it does not fall over and for the frequency of the back-and-forth motion. This matches prompt0 §1. This source is *Used* because it fixes the problem being solved; nothing else in the derivation rests on a source.

Consulted

Background, orientation, and post-hoc comparison. The result does not depend on any of these.

2. **The published solution**, David Morin, Solution Week 67 (12/22/03), “Inverted pendulum”, sol67.pdf, <https://www.physics.harvard.edu/sites/g/files/omnuum6476/files/physics/files/sol67.pdf>, eqs. (3), (4), (6), (10). {ver: tool} Retrieval note: fetched after §3 and §4 were complete. The fetched text gives the Lagrangian $L = \frac{1}{2}m(\ell^2\dot{\theta}^2 + \dot{y}^2 - 2\ell\dot{y}\dot{\theta}\sin\theta) - mg(y + \ell\cos\theta)$ as eq. (3), the equation of motion $\ell\ddot{\theta} - \ddot{y}\sin\theta = g\sin\theta$ as eq. (4), the small-angle form $\ddot{\theta} + \theta(a\omega^2\cos\omega t - \omega_0^2) = 0$ with $\omega_0 = \sqrt{g/\ell}$ and $a = A/\ell$ as eq. (6), and $\Omega = \sqrt{a^2\omega^2/2 - g/\ell}$ as eq. (10), together with the condition $a\omega > \sqrt{2}\omega_0$ and the limit $\Omega \approx a\omega/\sqrt{2}$. It also states that θ is measured from the upward vertical. These agree with (10), (15), (17), (37) and (41) here. Listed as *Consulted* rather than *Used* because the result was already derived and verified before this was opened; removing it would change nothing in §3, §4 or §5, only §6.
3. **L. D. Landau and E. M. Lifshitz**, *Mechanics*, 3rd ed., Course of Theoretical Physics vol. 1, §30 “Motion in a rapidly oscillating field”. {ver: no} I did not retrieve the book, and I do not claim an equation number. What I can report is that a secondary source I did retrieve (item 4) attributes the effective-potential treatment of this exact system to *Mechanics* §30, which is consistent with my recollection but is not the same as having read the section. Per CONVENTIONS.md §1, the method itself is tagged [R] in the risk register rather than [S], and nothing rests on it: §3.3–§3.5 derives the averaging from the equation of motion without importing a formula.

4. **Wikipedia**, “Kapitza’s pendulum”, https://en.wikipedia.org/wiki/Kapitza%27s_pendulum. {ver: tool}, for the Landau attribution only. Retrieval note: fetched twice, including via ?action=raw. The rendering returned to me had the mathematics stripped or mangled — it reported a stability condition “ $a^2\omega^4 > 4gl$ ”, which is dimensionally inconsistent for a pivot amplitude a — so I have taken nothing quantitative from it and do not treat it as confirming any formula. The one thing that came through cleanly is the sentence attributing the effective potential for the slow motion to *Mechanics* §30 of Landau’s course.
5. **E. I. Butikov**, “An improved criterion for Kapitza’s pendulum stability”. {ver: no} Located in a search but not retrieved and not read. Named here only because it is the obvious place to look for the $O(\varepsilon^2)$ threshold correction observed numerically in §4.4, and it would be dishonest to present that observation as unexplored territory without saying that such work exists.
6. **G. E. Astrakharchik and N. A. Astrakharchik**, “Numerical study of Kapitza pendulum”, arXiv:1103.5981. {ver: tool} for the abstract page only; <https://arxiv.org/abs/1103.5981>. Retrieval note: the abstract was fetched and describes numerical simulation of the driven pendulum validated against exact limiting cases. No criterion or formula was extracted, and nothing from it is used.
7. **E. Folkers**, *Floquet’s Theorem*, Bachelor’s project in Mathematics, Faculty of Science and Engineering, July 2018; Theorem 2.8 §2.4 (Floquet’s theorem), Lemma 3.12 §3.4 ($\det C = 1$ for Hill’s equation), §3.4 cases 1–3 (the trace criterion). https://fse.studenttheses.ub.rug.nl/17640/1/bMATH_2018_FolkersE.pdf {ver: tool} Retrieved 2026-09-14, in response to finding F1.2. Retrieval note: the document was fetched and returns, at those locators, the Floquet normal form $X(t) = Q(t)e^{Bt}$ with $C = X(T) = e^{BT}$; “ $\det C = 1$ ” for Hill’s equation with a proof by constancy of the Wronskian; and the three cases $|\operatorname{tr} C| > 2$ unstable, < 2 stable but not asymptotically, $= 2$ depending on semisimplicity. These agree with (58) and (59) here. The host is the University of Groningen thesis repository, identified from the URL rather than from the document’s own title page, which the fetched rendering did not name — so the institution is reported with that caveat rather than asserted. Listed as *Consulted*: §4.4 derives $\det M = 1$ and the trace criterion from scratch, so removing this source would change the tag on one sentence and nothing else.

On the round-1 retrieval of the published solution, and §6 of CONVENTIONS.md. Retrieving sol167.pdf during round 1 spent it as an independent check. It is now inside the loop: rounds 2 and later are downstream of it and cannot also be validated by it. meta.yaml records this as ground_truth.arrived: mid-loop. I note it again here because the tension is structural rather than a slip — prompt0 §6.3 requires the solver to retrieve and confirm its *Used* sources, and for a problem whose source is a problem sheet with a solution sheet beside it, obeying §6.3 burns the ground truth. The clean fix is for the solution sheet to be retrieved by the auditor rather than the solver, or after the loop closes.

9. Open questions

1. **The $O(\varepsilon^2)$ correction to the threshold.** N4 measures $G_c = (\varepsilon^2/2)(1 + c\varepsilon^2 + \dots)$ with $c = -0.4374$ over a factor of 32 in ε , which is $-7/16$ to the precision available. I have not derived it and I have not sourced it. Method I would use: second-order averaging on (17), or equivalently a perturbative expansion of the lowest stability boundary of the Hill

equation (57) in powers of ε , and then compare against Butikov’s improved criterion (§8 item 5). What would settle it is a symbolic calculation producing $-7/16$ exactly, not a fit.

2. **Where the frequency correction changes sign.** New from round 2. Check N8 finds the $O(\varepsilon^2)$ correction to Ω negative near threshold and positive far above it, crossing zero at $\mu \approx 0.7$ for $\varepsilon = 0.1$, where $\mu = 2gl/(A^2\omega^2) = \cos \Theta_c$. Whether μ_* is a pure number independent of ε , and what it is, is not answered here — the scan was run at one ε . Method: the same second-order averaging as item 1 would produce Ω^2 to $O(\varepsilon^4)$, and μ_* is where its correction term vanishes; the numerical side is a two-parameter scan of N8, which is cheap. The two must agree, and the sign structure is constrained at one end by N4’s boundary.
3. **The basin boundary beyond leading order.** N5 verifies Θ_c to $O(\varepsilon)$ once the ripple offset is accounted for, with a clean $O(\varepsilon^2)$ residual of coefficient -0.58 . Whether that coefficient has a closed form, and whether the true basin is exactly the averaged well or a slightly deformed version of it, is untested. Method: a stroboscopic (Poincaré) map at the drive period, with the boundary located as the stable manifold of the period-1 saddle rather than by a finite-window escape criterion.
4. **How long is “stays up”?** Everything here is either linear stability or a finite-window integration. The averaged system is conservative, so it predicts the pendulum stays up forever, but the exact system is not, and slow diffusion in the neglected $O(\varepsilon^2)$ terms could in principle eject a trajectory over very long times. Method: long integrations at fixed ε with an adiabatic-invariant diagnostic, or Nekhoroshev-type estimates.
5. **The upper stability boundary.** At larger ε the inverted position is destabilized again by parametric resonance. The scan in §4.4 stays well below it. Method: extend the Floquet scan in ε at fixed G and map the full tongue structure.
6. **Physical corrections.** A stick of finite mass, a pivot drive with a horizontal component, and linear damping each change the threshold in a way that is straightforward to work out and worth doing, since they are what separates the formula from a real demonstration. Damping in particular enlarges the stable region in ω while shrinking the basin, which is a nice sign-of-the-effect question to pose before computing.

Provenance

One row per tagged claim, built from the tags in the text. Generated from `sources/provenance.md`.

One row per tagged claim in `answers/answer-2.md`, built from the tags in the text rather than from the Sources section. This is the file that answers “did the model derive this or recall it”.

Tag meanings are in `templates/CONVENTIONS.md` §1. “Load-bearing” means the final result in `final/final.md` §Result changes if the claim is wrong.

Equation numbers are those of `answers/answer-2.md`, which `final/final.md` preserves.

Summary

Tag	Count	Meaning
[D]	23	Derived in the document from premises stated in it
[A]	9	Assumption, convention or ansatz

Tag	Count	Meaning
[N]	8 checks + 3 register rows	Numerical or symbolic computation
[V]	1	Checked against a source retrieved this session
[R]	2	Recalled from training, no locator confirmed
[S]	0	— none. Nothing is invoked as a standard result without either deriving it or marking it [R].
[?]	1	Solver not confident

Nothing load-bearing is [R], [?] or {ver: no}. The two [R] entries are attributions, not steps: removing both leaves every equation in §3 intact.

Derived claims — [D]

Eq. / §	Claim	Source	Locator	Verified	Load-bearing
§2.3	$\theta \equiv 0$ and $\theta \equiv \pi$ are exact solutions of (16) for all A, ω	this document	—	N1, N2	yes
(10)	The t -only group in L is annihilated by ∂_θ and $\partial_{\dot{\theta}}$ and may be dropped	this document	—	N1	yes
(15)	The $\dot{y}_p \dot{\theta} \cos \theta$ terms cancel identically; only the pivot's acceleration survives	this document	—	N1	yes
(16)	Equation of motion $l\ddot{\theta} = (g - A\omega^2 \cos \omega t) \sin \theta$	this document	—	N1, N2	yes
(17)	(15) is the pendulum equation with $g \rightarrow g_{\text{eff}}(t) = g + \ddot{y}_p$	this document	—	N2	no (consistency re-mark)
(17)	$A = 0$ reduces to $l\ddot{\theta} = g \sin \theta$, growth $e^{\omega_0 t}$	this document	—	N1	no
(20a)	$\mu = 2gl/(A^2\omega^2) = \cos \Theta_c$; stability is $\mu < 1$	this document	—	N4	yes
§3.3	Two blocks separated by exactly one factor of ε	this document	—	N3, N8	yes
§3.3	All three slow terms are $O(\varepsilon^2\omega^2)$; none may be dropped	this document	—	N4	yes
§3.3	Ordering inside the slow block degenerates at $\mu = 1$	this document	—	N4, §3.8	no
(24)	$c_1 = c_2 = 0$ in the fast solution	this document	—	N6	yes
(24)	$\xi = (A/l) \sin \Theta \cos \omega t$	this document	—	N6	yes
(27)– (28)	$\langle \cos \omega t \xi \rangle = \frac{1}{2}\varepsilon \sin \Theta$; the averaged equation	this document	—	N3, N8	yes
(32)	The added term of U_{eff} equals the time-averaged kinetic energy of the fast motion	this document	—	algebraic identity, §3.5	no (internal cross-check)
(38)	$W''(\pi) > 0$ always; the hanging position is never destabilised at this order	this document	—	self-check 3	no
(39)	$W''(\Theta_c) < 0$: the barrier is a maximum	this document	—	N5	yes (for Θ_c)

Eq. / §	Claim	Source	Locator	Verified	Load-bearing
(43)– (44)	$\Omega/\omega < \varepsilon/\sqrt{2}$, so the timescale separation follows from $A \ll l$	this document	—	N3, N8	yes (non-circularity)
(46)	$\beta < 0$ always; at threshold W has a quartic maximum, so (37) is strict	this document	—	N4, fig. 3	yes
(53)	Secular removal gives $c_1 = \frac{3}{4}\alpha a^2$	this document	—	N7	no
(56)	$T(a)/T(0) - 1 = a^2(\frac{1}{4} + 3\omega_0^2/16\Omega^2)$, always positive	this document	—	N7, N3 col. (c)	no
§4.3	The analytic size of N3’s column (c) is (56)	this document	—	N7	no
(58)	$\det M = 1$ exactly, by constancy of the Wronskian	this document	—	printed as $\det M - 1 \sim 10^{-15}$	yes (for N4, N8)
(59)	Linear stability of $\theta = 0$ is exactly $ \operatorname{tr} M < 2$	this document	—	N4 $\varepsilon = 0$ block	yes (for N4, N8)
§4.4	$\lambda_{\pm} = e^{\pm i\nu}$ with $2\cos\nu = \operatorname{tr} M$ defines the Floquet phase	this document	—	N8	yes (for N8)

Verified against a retrieved source — [V]

Eq. / §	Claim	Source	Locator	Verified	Load-bearing
§4.4, be- fore (59)	Floquet’s theorem: one matrix M advances the state over every period	E. Folkers, <i>Floquet’s Theorem</i> , 2018	Theorem 2.8 §2.4; Lemma 3.12 §3.4; §3.4 cases 1–3	{ ver: tool }, retrieved 2026-09-14; retrieval note in §8 item 7	no — §4.4 de- rives $\det M = 1$ and the trace crite- rion inde- pen- dently

Recalled — [R]

Eq. / §	Claim	Source	Locator	Verified	Load-bearing
§8 item 3, reg- is- ter R1	That the effective-potential method for a rapidly oscillating field is Landau & Lifshitz, <i>Mechanics</i> , §30	Landau & Lifshitz, <i>Mechanics</i> , 3rd ed.	§30 claimed, no equation number claimed	{ ver : no } — book not retrieved	no . §3.3–§3.5 derives the averaging from the equation of motion and imports no formula. no for the physics; yes for the applicability of Floquet theory. The coefficient of (57) is entire in τ .
§4.4, reg- is- ter R3	Existence and uniqueness for linear ODEs with continuous coefficients	—	none	not verified	

Both appear in the Risk register, so `no_unflagged_recall` holds. The [R] in §3.9 and the [S] in §8 are references to the tags themselves in prose, not claims: §3.9 says the relation was [R] in round 1 and is now derived, and §8 says an unconfirmable [S] must be downgraded.

Assumptions — [A]

Eq. / §	Assumption	What it buys	What breaks without it	Load-bearing
§2.1	θ measured from the upward vertical	reads naturally for the inverted case	nothing; relabel $\theta \rightarrow \pi - \theta$	no
(18)	Two-timescale split $\theta = \Theta + \xi$, $\langle \xi \rangle = 0$	an autonomous slow equation	all of §3.3–§3.6; the threshold survives via N4	yes
(20)	Truncating $\sin(\Theta + \xi)$ at first order in ξ	linearity in ξ	relative error $O(\varepsilon^2)$, measured by N3	yes

Eq. / §	Assumption	What it buys	What breaks without it	Load- bearing
(21)	Θ held fixed over one drive period	closed-form ξ	relative error $O(\varepsilon^2)$, measured by N3	yes
(49)	$\alpha a^2/\Omega^2$ small	closed-form amplitude correction	(56) only; no result in §5	no
§5 item 5	Massless rigid stick, point mass, frictionless pivot, planar motion, uniform g , exactly sinusoidal and exactly vertical drive	the posed problem	a massive stick replaces l by $I/(ml_{\text{cm}})$; damping shrinks the basin; a horizontal component breaks the $A \rightarrow -A$ symmetry	yes, but all are given in the problem statement
N2 code	Comparison tolerance 10^{-6} rad at <code>rtol=1e-11</code> ; refinement study on failure	a criterion that can fail	stated in §4.2, including the failure it produced	—
N5 code	Escape criterion $ \theta > 2.5$ rad within 25 slow periods	a decidable test	stated in §4.5; can only overestimate the basin	—
N7 code	$T(0)$ taken as $2\pi/\Omega$, the exact $a \rightarrow 0$ limit	a reference period	stated in the script header	—

No assumption lives only inside a `simplify` call or a solver option. Every assumption inside the code is restated as an [A] in the text, which is the `code_not_circular` requirement.

Computations — [N]

Check	Tests	File	Non-circular because	Mutants rejected
N1	(16) from the geometry, symbolically	<code>check1_eom_symbolic.py</code>	inputs are $x(t), y(t)$ and the definitions of T, V , not any line of §3.2	sign-flipped drive
N2	(16) against Newton + constraint force	<code>check2_newton_vs_lagrange.py</code>	no Lagrangian, no generalised coordinate	sign-flipped drive, 10^9 separation
N3	(41) from the exact nonlinear equation	<code>check3_slow_frequency.py</code>	integrates (16); Ω enters only at comparison time	$\times 2, \nabla \cdot 2, +g/l$
N4	(37) by exact Floquet theory	<code>check4_floquet_boundary.py</code>	no averaging anywhere; $\varepsilon = 0$ block checked against a closed form	$\varepsilon^2, \varepsilon^2/4, \varepsilon/2$
N5	(34), the barrier angle	<code>check5_basin_and_ripple.py</code>	bisects on the exact equation's behaviour	sub-threshold control holds nothing
N6	(24), the ripple envelope	<code>check5_basin_and_ripple.py</code>	ripple extracted from the exact solution by moving average	constant-amplitude ripple

Check Tests	File	Non-circular because	Mutants rejected
N7 (55)–(56) on the bare Duffing equation	<code>check6_duffing_shift.py</code>	no pendulum in the script at all	1/8, 3/4, sign flip
N8 (41) from the Floquet phase	<code>check7_floquet_frequency.py</code>	no averaging and no period fitting	$\times 2$, $\nabla \cdot 2$, $+g/l$

Three [N] rows in the Risk register are not checks but limitations of them: integrator tolerance (N2 set 8), N5’s finite window, and the shared `monodromy` routine between N4 and N8.

Low confidence — [?]

Eq. / §	Claim	Status
§4.4, reg- ister ?1	$G_c = (\varepsilon^2/2)(1 + c\varepsilon^2)$ with $c \rightarrow -0.4374 \approx -7/16$	Numerical observation over a factor of 32 in ε . Not derived, no source retrieved. Not used in §5. Queued as open question 1.

Independence record

The file to read first when revisiting this problem. It is the honest account of what the solver knew going in, and where the derivation could have leaned on it.

Assembled by `/log` from `answers/answer-2.md`: §1 (Prior-knowledge declaration) and §6 (Convergence check), both verbatim, plus the notes below.

Solver: Claude Opus 5, effort high (user-reported, from `prompts/prompt0.md` §0). Auditor: Gemini 3.1 Pro Avanzado, extended-reasoning option, effort unknown (user-reported).

1. Prior-knowledge declaration

Do I recognize this problem, and from where? Yes, immediately, and in two ways. It is the Kapitza pendulum — dynamic stabilization of the inverted position by rapid vertical vibration of the support — which I know as a standard worked example in the classical-mechanics literature. I also recognized the specific wording as a Harvard “Problem of the Week” sheet by David Morin, which `prompt0` §1 states outright. I did not open either source before deriving; see the ordering note below.

What result do I believe is the standard one? Written out before deriving anything, from recall alone:

- The pendulum does not fall because the fast shaking of the pivot, averaged over a drive period, contributes a term to an effective potential that has a minimum at the inverted position. The size of that term goes like the square of the amplitude of the fast wiggle, which itself is proportional to the sine of the angle from the vertical, so the extra potential is proportional to $\sin^2$ of that angle — a hill at the horizontal, hence a well at both vertical positions.
- The inverted position is stable when $A^2\omega^2 > 2gl$.
- The slow oscillation frequency is $\Omega = \sqrt{\frac{A^2\omega^2}{2l^2} - \frac{g}{l}}$, reducing to $\Omega \approx A\omega/(\sqrt{2}l)$ when $A\omega \gg \sqrt{gl}$.

What is the standard route, and where is it presented? The route I recall is Kapitza’s separation of the motion into a slow part and a fast part, with the fast part solved first and then averaged out, producing an effective potential for the slow part. I recall this being presented in Landau and Lifshitz, *Mechanics*, in the section on motion in a rapidly oscillating field, which I believe is §30 — recalled, not checked at the time of writing this section. I also recall the alternative route, which treats the linearized problem as a Mathieu equation and extracts the stability tongue.

What am I anchored on? On the three bullets above, which I wrote from memory and which the derivation below then had to reproduce. Section 3c of prompt0 says only “we expect some sort of equilibria”, which is too weak to anchor anything, though it is worth sharpening and I do so in §5. The honest statement is that I knew the target before I started, so the value of this document lies in whether each step is forced, and in whether the checks in §4 could have failed. I have tried to make both auditable: every numerical check in §4 is run not only on my claim but on deliberately mutated versions of it, and the mutants are reported alongside.

Ordering note, which matters for the audit. The derivation in §3 and every computation in §4 were completed and their outputs recorded before any source was retrieved. The retrieval in §8 happened afterwards, for the sole purpose of resolving prompt0’s general reference into a locator and checking it, as §6.3 requires. The comparison with what was retrieved is in §6. I mention this because the previous problem in this repository failed precisely by retrieving a published solution instead of deriving one, and the ordering is the only thing that distinguishes the two cases from the outside.

Perturbation test

Not run. `templates/audit-prompt.md` pass 8 and category C of the exploration prompt both call for it, and `claude/workflow-feedback.md` §3 calls it the sharpest test in the system. Neither the round-1 audit nor any later step performed one for this problem.

What it would have looked like here: change one thing so that the memorised answer becomes wrong, and see whether the derivation tracks the change or breaks. Three candidates, in increasing order of how much they would have told us.

1. Drive the pivot **horizontally** instead of vertically, $x_p(t) = A \cos \omega t$. The effective potential picks up a $\cos^2$ rather than a $\sin^2$ dependence, so the stabilised positions move to the horizontal rather than the vertical. A derivation that had memorised “ $\sin^2 \Theta$ ” would produce the wrong equilibria.
2. Replace the point mass by a **uniform rod** pivoted at one end. Then l is replaced by $I/(m l_{\text{cm}}) = \frac{2}{3}l$ in some places and not others, and the threshold and the frequency shift by different factors. A reconstruction would almost certainly rescale both the same way.
3. Drive at $y_p = A \cos \omega t + B \cos 2\omega t$. The averaging then has two fast components and the $\langle \cos^2 \rangle = \frac{1}{2}$ step acquires a second term; the answer is no longer a single $A^2 \omega^2$.

Recorded as not-run rather than as a gap to be filled retroactively: running it now, after the answer and the audit are both on the record, measures something different from running it inside the loop.

What the audit found about independence

Round 1, pass 8 — independence audit: **Pass**. “The timescale separation and balancing of orders is non-circular and properly motivated.” The auditor did not identify any anchoring step beyond the two the solver self-reported in §6.

Anchoring steps self-reported by the solver: 5 (listed in §6 below, items 1–5). Anchoring steps found by the auditor and not self-reported: 0.

That asymmetry is worth noticing rather than celebrating. An auditor that finds nothing the solver did not already confess has not independently probed the anchoring; it has read the confession and agreed with it. The blind-solve stage (`templates/blind-solve-prompt.md`) exists for exactly this and was not run for P002.

6. Convergence check

Does the derivation reproduce the declared result? Yes, in all three parts. §1 declared $A^2\omega^2 > 2gl$ and $\Omega = \sqrt{A^2\omega^2/(2l^2) - g/l}$ and the $\sin^2$ effective potential; (37), (41) and (31) are those. There is nothing to reconcile and no disagreement to report.

It also agrees with the published solution, retrieved after the fact (§8): that solution’s equation (4) is $l\ddot{\theta} - \dot{y}\sin\theta = g\sin\theta$, which is (15) here, and its equation (10) is $\Omega = \sqrt{a^2\omega^2/2 - g/l}$ with $a = A/l$, which is (41) here. The routes differ: the published solution linearizes first and works with the resulting Mathieu-type equation and an envelope ansatz, while §3 keeps the nonlinearity and constructs an effective potential. The effective-potential route yields Θ_c and the $\Theta = \pi$ results as by-products, which the linearized route does not reach.

It agrees, third, with a computation performed by a party that did not write this derivation: the round-1 auditor extracted the slow frequency from the Floquet phase of the linearized equation and found it consistent with (41) to $O(\varepsilon^2)$. That is the only check in this document whose method was not chosen by me, which makes it worth more than its arithmetic. I reproduced it (§4.9), corrected the sign of its reported deviation, and extended it into a sweep. Its real value turned out not to be the single point at all but the constraint it places on N4: the frequency it measures must vanish exactly where N4’s independent bisection puts the boundary, and it does.

Steps a person who did not know the target could not have chosen as I did. Listed even where I believe each is justified.

1. **The split $\theta = \Theta + \xi$ (18).** This is the method, and it is not forced by the problem. Someone meeting this cold could arrive at it by noticing that the equation contains exactly one imposed fast timescale, but they could equally linearize and reach for Floquet theory, as the published solution effectively does. I knew the method. Mitigation: N4 verifies the threshold by the other route entirely, so the answer does not depend on this choice being the right one.
2. **Dropping c_1 and c_2 in (23).** Motivated — $c_1 t$ contradicts boundedness, c_2 is already called Θ — but a reader who did not know that the answer contains no secular term might have hesitated. I regard this as forced rather than chosen, because the assumption being used to derive ξ is the same assumption that excludes c_1 .
3. **Keeping $\varepsilon\omega^2 \cos(\omega t) \xi \cos \Theta$ while $\langle \ddot{\xi} \rangle$ vanishes.** This is the crux, and it is the one step where knowing the answer could substitute for an argument. It is forced by the order count in §3.3: both that term and $\ddot{\Theta}$ are $O(\varepsilon^2\omega^2)$, whereas $\langle \ddot{\xi} \rangle$ is exactly zero by periodicity rather than small. If I had dropped it, the averaged equation would have had no stabilizing term at all and $\Theta = 0$ would never be stable — the derivation would have failed loudly, not quietly.
4. **Holding Θ fixed across a drive period** in (21) and in (25)–(27). This is the leading order of a systematic expansion and its error is $O(\varepsilon^2)$; the point of N3 is that this error was not asserted but measured, and came out at the predicted order with a clean coefficient.
5. **Nothing was inserted for later convenience.** The factor $\frac{1}{2}$ in (28) is $\langle \cos^2 \rangle$ and enters at (27); the factor $\frac{1}{4}$ in (29) is half of it, from writing $\sin \Theta \cos \Theta$ as a derivative. Neither was available to be tuned.

Would the derivation still run if the standard result were slightly different? No, and the places it would break are identifiable.

- **A different numerical factor.** The $\frac{1}{2}$ comes from $\langle \cos^2 \omega t \rangle$ at (27) and nowhere else. Had the target been $A^2 \omega^2 > gl$ or $> 4gl$, there is no step in §3 that could have been bent to produce it; I would have had a derivation disagreeing with my recollection and would have had to report both, per the protocol. I would have noticed, because the mutation blocks in N3 and N4 reject those factors outright.
- **A different power of ω .** The threshold's ω^2 and the fact that ξ is ω -independent both trace to the exact cancellation at (21)–(24): $\ddot{y}_p$ brings down ω^2 and the double integration removes it. That cancellation is visible and unforced. If the standard result had $\Omega \propto \sqrt{\omega}$, say, the derivation would have broken at (24), where there is simply nowhere for a half-power to come from.
- **A different power of A .** Ruled out independently by the symmetry argument in §4.6 item 4: results must be even in A .

The most honest statement of residual risk is this. I knew the answer before I started, and no amount of prose can make that not so. What can be checked from outside is whether the checks in §4 could have failed, and they could: N2 did fail on its first run and the failure is reported; every claim is run against mutants that are rejected; and the two strongest checks, N3 and N4, are convergence tests against exact routes that share no algebra with §3.3–§3.6.

Audit history

One round. The solver produced `answers/answer-1.md`; the auditor returned a nine-pass report with verdict `ACCEPT-WITH-FIXES` and two findings, both `MINOR`; the solver produced `answers/answer-2.md`, from which this document is derived.

Auditor: Gemini 3.1 Pro Avanzado with the extended-reasoning option, effort level unknown. Both are user-reported; the report carried no run header, and neither value was guessed.

Finding F1.1, `MINOR`, against §3.3: the size-counting table asserted $\omega_0^2 \ll \varepsilon^2 \omega^2$ near threshold, which is false, since at the boundary $\varepsilon^2 \omega^2 = 2\omega_0^2$ exactly. Accepted in full. §3.3 was rewritten around $\mu = 2gl/(A^2 \omega^2)$, the same combination that appears as $\cos \Theta_c$ in eq. (34), and the corrected counting now says what it should: all three slow terms are of the same order and none may be dropped. The false line was in the justification and not in a step, and nothing downstream rested on it; had it been acted on, the averaged equation would have lost gravity and there would have been no threshold at all.

Finding F1.2, `MINOR`, against §4.3 and §4.4: standard results invoked without the provenance tags `CONVENTIONS §1` requires. Accepted, and answered by deriving them rather than tagging them. The amplitude–frequency relation for a quartic well is now derived by Lindstedt–Poincaré in §3.9 and tested on its own in check N7; $\det M = 1$ and the $|\operatorname{tr} M| < 2$ criterion are derived in §4.4. Risk-register entry R2 was retired and R3 added for the one item that remains recalled.

No finding was rebutted. Both were correct, so `rebuttals_upheld` is zero for this problem and carries no information about whether the solver would have disagreed; `claude/workflow-feedback.md §5` is the reason that count is tracked at all.

The solver raised one item against the audit rather than against a finding. The audit's pass 6 reports `Exact: 4.2764` from a script it quotes; run verbatim, that script prints `Exact: 4.2867`, and five integrators agree on $\operatorname{tr} M = 1.967843917562$. The audit's conclusion survives — the Floquet frequency does agree with the closed form to $O(\varepsilon^2)$ — but the sign of the deviation is reversed. The audit text is kept verbatim in `audits/audit-report-1.md` with the

correction appended as a note, since a wrong number in an audit report is an accurate record of what the auditor said. The method itself was adopted as check N8, swept in ε , and used to test N4 and N8 against each other.

`answers/answer-2.md` was not itself audited. The stopping rule in CONVENTIONS §6 was met at round 1, which returned no BLOCKER and no MAJOR, and ACCEPT-WITH-FIXES means one further pass closes the MINORS; that pass is answer-2. Its new material — §3.9, check N7 and check N8 — has therefore never been read by an auditor. Recorded here rather than resolved.

Code

Every script as executed. Outputs are quoted verbatim in §4 above; the files themselves are `assets/code/out_check*.txt`, each paired with a `.coverage.md` stating what that check would and would not have caught.

`check1_eom_symbolic.py`

```
#!/usr/bin/env python3
# check1_eom_symbolic.py -- P002, check N1
#
# Purpose: derive the equation of motion from the GEOMETRY of the problem
# (the Cartesian position of the bob) rather than from any expression that
# appears in the hand derivation, and compare the result with the claimed
# equation of motion
#
#      l*thetadd = (g - A*w**2*cos(w*t)) * sin(theta).          (claim)
#
# Nothing below reuses an intermediate result of the hand derivation: the only
# inputs are x(t), y(t), the kinetic and potential energy definitions, and the
# Euler-Lagrange operator.
#
# Environment (pinned):
#   python 3.11.15, sympy 1.14.0
# No randomness is used in this script, so no seed is needed.

import sympy as sp

t = sp.symbols('t', real=True)
m, l, g, A, w = sp.symbols('m l g A omega', positive=True)
th = sp.Function('theta', real=True)(t)

# --- geometry, straight from the problem statement -----
# theta is measured from the UPWARD vertical; the pivot sits at (0, y_p(t)).
y_p = A * sp.cos(w * t)
x = l * sp.sin(th)
y = y_p + l * sp.cos(th)

xd = sp.diff(x, t)
yd = sp.diff(y, t)

T = sp.Rational(1, 2) * m * (xd**2 + yd**2)
V = m * g * y
L = sp.simplify(T - V)
print("L =", sp.simplify(sp.expand_trig(L)))

# --- Euler-Lagrange -----
EL = sp.diff(sp.diff(L, sp.diff(th, t)), t) - sp.diff(L, th)
EL = sp.simplify(sp.expand(EL))
print("Euler-Lagrange expression (should equal 0):")
```

```

print(sp.simplify(EL))

# Solve for theta''
thdd = sp.solve(sp.Eq(EL, 0), sp.diff(th, t, 2))
print("theta'' solved from Euler-Lagrange:")
for s in thdd:
    print("    ", sp.simplify(s))

claim = (g - A * w**2 * sp.cos(w * t)) * sp.sin(th) / l
print("claimed theta'' =", claim)

assert len(thdd) == 1, "unexpected number of solutions"
resid = sp.simplify(sp.expand_trig(sp.simplify(thdd[0] - claim)))
print("residual (solved - claimed) =", resid)
print("CHECK N1:", "PASS" if resid == 0 else "FAIL")

# --- a deliberately wrong claim, to show the check can fail -----
# If the check cannot reject a wrong equation it is worth nothing, so run it
# against one. The mutant has the driving term with the wrong sign.
mutant = (g + A * w**2 * sp.cos(w * t)) * sp.sin(th) / l
resid_mutant = sp.simplify(sp.expand_trig(sp.simplify(thdd[0] - mutant)))
print("residual against sign-flipped mutant =", resid_mutant)
print("MUTANT REJECTED:", "yes" if resid_mutant != 0 else "no -- the check is
    vacuous")

# --- the A -> 0 limit -----
print("A -> 0 limit of theta'':", sp.simplify(thdd[0].subs(A, 0)))

```

check2_newton_vs_lagrange.py

```

#!/usr/bin/env python3
# check2_newton_vs_lagrange.py -- P002, check N2 (second version; see the
# note at the bottom of this header for what changed after the first version
# failed, and why)
#
# Purpose: test the equation of motion
#
#  $l \cdot \text{thetadd} = (g - A \cdot w^2 \cdot \cos(w \cdot t)) \cdot \sin(\text{theta})$  (claim)
#
# against a SECOND, independently formulated dynamics: Newton's second law in
# Cartesian coordinates for the bob, with the rigid stick represented by a
# constraint force along the stick whose magnitude is fixed by differentiating
# the constraint  $|r - p(t)| = l$  twice. No Lagrangian, no generalised
# coordinate, and no expression lifted from the hand derivation.
#
# Let  $u = r - p(t)$  be the bob position relative to the pivot,  $|u| = l$ .
#  $m \cdot \text{rdd} = -m \cdot g \cdot \text{jhat} + (\lambda / l) \cdot u$  (force along the stick)
#  $\text{udd} = \text{rdd} - \text{pdd}$ 
#  $u \cdot u = l^2 \Rightarrow u \cdot \text{ud} = 0 \Rightarrow u \cdot \text{udd} + |\text{ud}|^2 = 0$ 
# which gives  $k := \lambda / (m \cdot l) = ((g + \text{pdd}_y) \cdot u_y - |\text{ud}|^2) / l^2$ 
# and  $\text{udd} = (k \cdot u_x, -(g + \text{pdd}_y) + k \cdot u_y)$ .
#
# Environment (pinned):
# python 3.11.15, numpy 2.4.4, scipy 1.17.1
# Parameter sets are drawn with numpy PCG64 seeded at 20260913. Every set
# drawn is reported.
#
# WHAT CHANGED AFTER THE FIRST RUN (reported in full in the answer): the first
# version applied a single fixed criterion,  $\max |d\text{theta}| < 1e-6$  over a 6 s
# window, to all eight sets. Set 8 failed it. Set 8 sits BELOW the stability
# threshold, so the pendulum falls and then whirls, and in that regime the
# motion has a positive Lyapunov exponent: two numerically exact but distinct

```

```

# integrations of the same equations separate exponentially. The criterion was
# therefore wrong for that regime, not the physics. The change made was to the
# CHECK, not to the derivation: the fixed-tolerance criterion is kept, and any
# set that fails it is then subjected to a tolerance-refinement study. A
# discrepancy that falls when rtol is tightened -- in step with the independent
# witness  $|u| - l$ , which the theta-equation cannot drift in by construction --
# is integrator error. A discrepancy that does not fall is a real
# disagreement. The cost of this is stated in the coverage note.

import numpy as np
from scipy.integrate import solve_ivp

SEED = 20260913
rng = np.random.default_rng(SEED)
G = 9.81

def rhs_theta(t, s, g, l, A, w, sign=-1.0):
    """Scalar equation of motion in theta (the claim).
    sign=-1 is the claim; sign=+1 is a deliberately wrong mutant."""
    th, thd = s
    return [thd, (g + sign * A * w**2 * np.cos(w * t)) * np.sin(th) / l]

def rhs_cartesian(t, s, g, l, A, w):
    """Newton + constraint force, in Cartesian components of  $u = r - p(t)$ ."""
    ux, uy, vx, vy = s
    pdd_y = -A * w**2 * np.cos(w * t)
    k = ((g + pdd_y) * uy - (vx * vx + vy * vy)) / l**2
    return [vx, vy, k * ux, -(g + pdd_y) + k * uy]

def run_pair(g, l, A, w, th0, thd0, tmax, rtol=1e-11, n=2001, sign=-1.0):
    ts = np.linspace(0.0, tmax, n)
    kw = dict(t_eval=ts, rtol=rtol, atol=rtol / 10, method='DOP853',
              max_step=0.2 * 2 * np.pi / w)
    a = solve_ivp(rhs_theta, (0, tmax), [th0, thd0], args=(g, l, A, w, sign),
                  **kw)
    u0 = [l * np.sin(th0), l * np.cos(th0),
          l * thd0 * np.cos(th0), -l * thd0 * np.sin(th0)]
    b = solve_ivp(rhs_cartesian, (0, tmax), u0, args=(g, l, A, w), **kw)
    assert a.success and b.success, (a.message, b.message)
    th_b = np.arctan2(b.y[0], b.y[1])
    d = np.abs(np.unwrap(a.y[0]) - np.unwrap(th_b)) # unwrap: a 2pi wrap is
    drift = np.abs(np.hypot(b.y[0], b.y[1]) - l) / l # not a disagreement
    return d.max(), drift.max()

print("CHECK N2 -- scalar theta-equation vs Newton-with-constraint-force.")
print("Both integrations start from the same physical state.")
print("Primary criterion: max|dtheta| < 1e-6 rad over the window at rtol=1e-11.")
print()
print("A real disagreement would show as a discrepancy that does NOT fall when")
print()
print("rtol is tightened. Both integrators are DOP853, so a shared")
print("discretisation bias is not credible at these tolerances.")
print()
hdr = (f"{'set':>3} {'l':>6} {'A':>8} {'omega':>8} {'w/w_c':>6} {'th0':>8} "
      f"{'tmax':>6} {'max|dth|':>11} {'|u|drift':>10} verdict")
print(hdr)

sets, failed = [], []
for i in range(8):

```

```

l = float(rng.uniform(0.2, 1.5))
A = float(rng.uniform(0.005, 0.05) * l)          # A << l
wc = np.sqrt(2 * G * l) / A                      # averaged threshold
factor = float(rng.uniform(1.2, 2.5)) if i < 6 else float(rng.uniform(0.4,
0.8))
w = factor * wc
th0 = float(rng.uniform(-0.3, 0.3))
tmax = min(40 * 2 * np.pi / w * 5, 6.0)
d, drift = run_pair(G, l, A, w, th0, 0.0, tmax)
ok = d < 1e-6
if not ok:
    failed.append((i + 1, l, A, w, th0, tmax))
sets.append((l, A, w, factor, th0, tmax))
print(f"{i+1:>3} {l:6.3f} {A:8.5f} {w:8.1f} {factor:6.2f} {th0:8.4f} "
      f"{tmax:6.2f} {d:11.3e} {drift:10.2e} {'PASS' if ok else 'FAIL'}")

print()
print(f"Primary criterion: {8-len(failed)}/8 sets passed.")

if failed:
    print()
    print("Tolerance-refinement study on the set(s) that failed.")
    print("If the discrepancy is integrator error it falls with rtol, and the")
    print("constraint residual |u|-1 falls with it. If the two formulations")
    print("actually disagree, neither falls.")
    for (idx, l, A, w, th0, tmax) in failed:
        print(f" set {idx}: l={l:.4f} A={A:.5f} omega={w:.1f} th0={th0:.4f}
              ")
        prev = None
        for rtol in [1e-8, 1e-10, 1e-12, 1e-13]:
            d, drift = run_pair(G, l, A, w, th0, 0.0, tmax, rtol=rtol, n=4001)
            trend = "" if prev is None else f" (x{d/prev:.2e})"
            prev = d
            print(f"      rtol={rtol:.0e} max|dth|={d:.3e} "
                  f"max||u|-1|/l={drift:.3e}{trend}")
            # and the same comparison restricted to the first 20 drive periods,
            # before exponential separation has had time to act
            tshort = 20 * 2 * np.pi / w
            d_s, drift_s = run_pair(G, l, A, w, th0, 0.0, tshort, rtol=1e-11)
            print(f"      first 20 drive periods (t < {tshort:.4f} s): "
                  f"max|dth|={d_s:.3e} -> {'PASS' if d_s < 1e-6 else 'FAIL'}")

# --- mutation test: can this check reject a wrong equation of motion? -----
print()
print("Mutation test (super-threshold set, where the primary criterion holds):")
)
l, A = 1.0, 0.02
w = 1.6 * np.sqrt(2 * G * l) / A
d_ok, _ = run_pair(G, l, A, w, 0.2, 0.0, 2.0, sign=-1.0)
d_mut, _ = run_pair(G, l, A, w, 0.2, 0.0, 2.0, sign=+1.0)
print(f" claim as derived : max|dtheta| = {d_ok:.3e}")
print(f" driving term sign flipped : max|dtheta| = {d_mut:.3e}")
print(" MUTANT REJECTED:", "yes" if d_mut > 1e-6 else "NO -- the check is
vacuous")

```

diag_set8.py

```

#!/usr/bin/env python3
# diag_set8.py -- diagnosis of the FAILING set 8 in check2.
# Question: is the disagreement a disagreement between the two FORMULATIONS,
# or is it the Cartesian integration losing the constraint |u| = l?
# Test: tighten rtol/atol and see whether max|dtheta| and the constraint drift

```

```

# fall together. If they do, it is integrator error, not physics.

import numpy as np
from scipy.integrate import solve_ivp

SEED = 20260913
rng = np.random.default_rng(SEED)
g = 9.81
sets = []
for i in range(8):
    l = float(rng.uniform(0.2, 1.5))
    A = float(rng.uniform(0.005, 0.05) * l)
    wc = np.sqrt(2 * g * l) / A
    factor = float(rng.uniform(1.2, 2.5)) if i < 6 else float(rng.uniform(0.4,
        0.8))
    w = factor * wc
    th0 = float(rng.uniform(-0.3, 0.3))
    sets.append((l, A, w, th0, min(40 * 2 * np.pi / w * 5, 6.0), factor))

for i, s in enumerate(sets, 1):
    print(i, [f"{v:.5g}" for v in s])
print()

l, A, w, th0, tmax, factor = sets[7]

def rhs_theta(t, s):
    th, thd = s
    return [thd, (g - A * w**2 * np.cos(w * t)) * np.sin(th) / l]

def rhs_cart(t, s):
    ux, uy, vx, vy = s
    pdd_y = -A * w**2 * np.cos(w * t)
    k = ((g + pdd_y) * uy - (vx * vx + vy * vy)) / l**2
    return [vx, vy, k * ux, -(g + pdd_y) + k * uy]

ts = np.linspace(0, tmax, 4001)
for tol in [1e-9, 1e-11, 1e-13, 1e-14]:
    a = solve_ivp(rhs_theta, (0, tmax), [th0, 0.0], t_eval=ts, rtol=tol,
        atol=tol / 10, method='DOP853', max_step=0.2 * 2 * np.pi / w)
    u0 = [l * np.sin(th0), l * np.cos(th0), 0.0, 0.0]
    b = solve_ivp(rhs_cart, (0, tmax), u0, t_eval=ts, rtol=tol, atol=tol / 10,
        method='DOP853', max_step=0.2 * 2 * np.pi / w)
    th_b = np.arctan2(b.y[0], b.y[1])
    d = np.abs(np.unwrap(a.y[0]) - np.unwrap(th_b))
    drift = np.abs(np.hypot(b.y[0], b.y[1]) - l) / l
    # first time the two curves separate by more than 1e-6
    idx = np.argmax(d > 1e-6) if np.any(d > 1e-6) else -1
    tsep = ts[idx] if idx >= 0 else np.inf
    print(f"rtol={tol:.0e} max|dth|={d.max():.3e} max drift={drift.max():.3e}
        "
        f"separation time={tsep:.3f} s (tmax={tmax:.2f})")

print()
print("Number of drive periods in the window:", tmax * w / (2 * np.pi))
print("factor omega/omega_crit =", factor, " -> below threshold, pendulum falls
    ")

```

check3_slow_frequency.py

```

#!/usr/bin/env python3
# check3_slow_frequency.py -- P002, check N3
#
# Purpose: test the headline result
#
#      Omega = sqrt( A^2*w^2/(2*l^2) - g/l )                                (claim)
#
# by integrating the EXACT equation of motion -- the one that comes out of the
# problem statement, with no averaging anywhere in the integrator -- and
# measuring the period of the slow back-and-forth motion from its zero
# crossings. The claimed Omega is used only at the end, to be compared
# against; it never enters the integration.
#
# Design of the sweep. The averaging is an expansion in  $\epsilon = A/l$ , so a single
# pass/fail at one tolerance says little. Instead  $\epsilon$  is swept downwards with
# Omega HELD FIXED ( $w$  is solved for at each  $\epsilon$ ), and the residual is required
# to fall like  $\epsilon^2$ , which is the order of the first neglected term. A wrong
# numerical factor inside Omega produces an  $O(1)$  residual that does not fall at
# all, which is what the mutation test at the bottom demonstrates.
#
# Two comparisons are reported per  $\epsilon$ :
# (a) measured period vs  $2\pi/\Omega$  -- the headline claim, harmonic;
# (b) measured period vs the period of the averaged equation integrated at
# the SAME initial amplitude -- this removes the anharmonicity of the
# effective potential, which is a real effect and not an error, and so
# isolates the error of the averaging step itself.
#
# Environment (pinned): python 3.11.15, numpy 2.4.4, scipy 1.17.1
# No random numbers are used; the sweep is deterministic.

import numpy as np
from scipy.integrate import solve_ivp

g, l = 9.81, 1.0
TH0 = 0.02          # small, so that the anharmonic shift is itself small
N_SLOW = 12         # slow periods integrated

def exact_rhs(t, s, A, w):
    th, thd = s
    return [thd, (g - A * w**2 * np.cos(w * t)) * np.sin(th) / l]

def averaged_rhs(t, s, A, w):
    TH, THd = s
    return [THd, (g / l) * np.sin(TH) - (A**2 * w**2 / (2 * l**2)) * np.sin(TH) * np.cos(TH)]

def zero_crossing_period(ts, th):
    """Half-period from a straight-line fit to the zero-crossing times.
    Returns (period, n_crossings, rms residual of the fit in seconds)."""
    s = np.sign(th)
    idx = np.where(s[:-1] * s[1:] < 0)[0]
    if len(idx) < 4:
        return np.nan, len(idx), np.nan
    t0, t1 = ts[idx], ts[idx + 1]
    y0, y1 = th[idx], th[idx + 1]
    tc = t0 - y0 * (t1 - t0) / (y1 - y0)          # linear interpolation
    k = np.arange(len(tc))
    slope, inter = np.polyfit(k, tc, 1)
    resid = np.sqrt(np.mean((tc - (slope * k + inter))**2))

```

```

    return 2 * slope, len(tc), resid

def measure(A, w, rhs, tmax):
    ts = np.linspace(0, tmax, 400001)
    sol = solve_ivp(rhs, (0, tmax), [TH0, 0.0], t_eval=ts, args=(A, w),
                    rtol=1e-11, atol=1e-13, method='DOP853',
                    max_step=2 * np.pi / w / 20)
    assert sol.success, sol.message
    return zero_crossing_period(ts, sol.y[0])

def omega_claim(A, w):
    val = A**2 * w**2 / (2 * l**2) - g / l
    return np.sqrt(val) if val > 0 else np.nan

print("CHECK N3 -- slow frequency measured from the exact equation of motion.")
print(f"g = {g}, l = {l}, theta(0) = {TH0} rad, thetadot(0) = 0.")
print("Omega is held at 6.000 rad/s while eps = A/l is swept; omega is solved")
print("for from the claim at each eps. If the claim is right the measured")
print("period converges on 2*pi/Omega like eps^2. A residual that plateaus,")
print("or that falls at the wrong order, is a failure.")
print()

OM = 6.0
T_harm = 2 * np.pi / OM

# Analytic size of the anharmonic shift, for comparison. Expanding the
# effective potential per unit m*l^2, W(TH) = (g/l) cos TH + (A^2 w^2/(4 l^2))
# sin^2 TH, about TH = 0 gives THdd = -Omega^2 TH - 4*beta*TH^3 with
# beta = g/(24 l) - A^2 w^2/(12 l^2) = -(Omega^2/6 + g/(8 l)). For
# xdd + Om^2 x + alpha x^3 = 0 with alpha = 4 beta, the amplitude-dependent
# frequency is Om(1 + 3 alpha a^2/(8 Om^2)), so the RELATIVE PERIOD shift is
# -3*alpha*a^2/(8*Om^2) = -3*beta*a^2/(2*Om^2). This is a property of the
# averaged system, not an error of the averaging.
beta = -(OM**2 / 6 + g / (8 * l))
anh_pred = -3 * beta * TH0**2 / (2 * OM**2)
print(f"Predicted anharmonic period shift at theta(0) = {TH0}: "
      f"{anh_pred:+.4e} (relative). This is an eps-independent floor under")
print("comparison (a) and must NOT be read as an error of the averaging.")
print()

print(f"{'eps=A/l':>9} {'A':>9} {'omega':>10} {'w/OM':>7} {'T_meas':>10} "
      f"{'(a)=T/Th-1':>12} {'(b)=T/Tav-1':>13} {'(c)=Tav/Th-1':>13} "
      f"{'(b)/eps^2':>10} {'ncross':>6} {'fitres':>9}")

rows = []
for eps in [0.1, 0.05, 0.025, 0.0125, 0.00625]:
    A = eps * l
    w = (np.sqrt(2.0) * l / A) * np.sqrt(OM**2 + g / l) # inverts the claim
    tmax = N_SLOW * 2 * np.pi / OM
    T_ex, nc, res = measure(A, w, exact_rhs, tmax)
    T_av, _, _ = measure(A, w, averaged_rhs, tmax)
    ra = T_ex / T_harm - 1 # signed
    rb = T_ex / T_av - 1 # signed: the averaging error
    rc = T_av / T_harm - 1 # signed: the anharmonic shift, measured
    rows.append((eps, ra, rb, rc))
    print(f"{eps:9.5f} {A:9.5f} {w:10.2f} {w/OM:7.1f} {T_ex:10.6f} "
          f"{ra:+12.3e} {rb:+13.3e} {rc:+13.3e} "
          f"{abs(rb)/eps**2:10.3f} {nc:6d} {res:9.2e}")

print()

```

```

print("Decomposition check: (a) should equal (b) + (c) + 0(second order).")
for eps, ra, rb, rc in rows:
    print(f"  eps={eps:8.5f}:  (a)={ra:+.4e}    (b)+(c)={rb+rc:+.4e}    "
          f"residual={ra-rb-rc:+.2e}")

print()
print("Observed convergence orders (log-log slopes between consecutive eps).")
print("Column (b) is the one that tests the averaging; (c) is eps-independent")
print("by construction and is shown to confirm that.")
for i in range(len(rows) - 1):
    e0, a0, b0, c0 = rows[i]
    e1, a1, b1, c1 = rows[i + 1]
    pb = np.log(abs(b0 / b1)) / np.log(e0 / e1)
    print(f"  eps {e0:.5f} -> {e1:.5f}:  order(b) = {pb:5.2f}    "
          f"(c) = {c0:+.3e} -> {c1:+.3e}")

print()
print("MUTATION TEST.  The same measurement, but the period is compared with")
print("mutant formulas.  Each mutant is used to solve for omega at eps = 0.02,"
      )
print("then the exact equation is integrated and the period measured.  A")
print("mutant that survives would mean the check cannot tell formulas apart.")
eps = 0.02
A = eps * l
mutants = {
    "claim    Omega^2 = A^2*w^2/(2l^2) - g/l": lambda: (np.sqrt(2.0) * l / A) *
        np.sqrt(OM**2 + g / l),
    "mutant    Omega^2 = A^2*w^2/(1l^2) - g/l": lambda: (1.0 * l / A) * np.sqrt(OM
        **2 + g / l),
    "mutant    Omega^2 = A^2*w^2/(4l^2) - g/l": lambda: (2.0 * l / A) * np.sqrt(OM
        **2 + g / l),
    "mutant    Omega^2 = A^2*w^2/(2l^2) + g/l": lambda: (np.sqrt(2.0) * l / A) *
        np.sqrt(max(OM**2 - g / l, 1e-9)),
}
for name, wfun in mutants.items():
    w = wfun()
    T_ex, nc, _ = measure(A, w, exact_rhs, N_SLOW * 2 * np.pi / OM)
    rel = abs(T_ex / (2 * np.pi / OM) - 1)
    print(f"  {name}:  omega = {w:9.2f}  T_meas = {T_ex:9.6f}    "
          f"rel err = {rel:9.3e}  -> {'consistent' if rel < 1e-2 else 'REJECTED'
          }")

```

check4_floquet_boundary.py

```

#!/usr/bin/env python3
# check4_floquet_boundary.py  --  P002, check  $N_4$ 
#
# Purpose: test the stability condition
#
#       $A^2 * w^2 > 2 * g * l$                                      (claim)
#
# against exact Floquet theory for the linearised equation of motion, which
# involves no averaging at all.
#
# Linearising the exact equation about  $\theta = 0$  and putting  $\tau = w*t$ ,
#
#       $\theta'' = (G - \text{eps} * \cos(\tau)) * \theta$  ,
#       $G = g/(l*w^2)$ ,       $\text{eps} = A/l$  ,      ' =  $d/d\tau$  .
#
# This is a Hill equation with period  $2\pi$  in  $\tau$ .  Its monodromy matrix  $M$  is
# built by integrating from the two initial conditions (1,0) and (0,1) over one
# period.  The equation has no damping, so  $\det M = 1$  exactly, and the origin is

```

```

# linearly stable iff  $|\text{tr } M| < 2$ . The claim, rewritten in these variables, is
#
# stable  $\Leftrightarrow \text{eps}^{2/2} > G$ , i.e. the boundary is  $G_c = \text{eps}^{2/2}$ .
#
# The test is not "does one point agree" but "does the exactly computed
# boundary  $G_c(\text{eps})$  approach  $\text{eps}^{2/2}$  as  $\text{eps} \rightarrow 0$ , and at what rate".
#
# Environment (pinned): python 3.11.15, numpy 2.4.4, scipy 1.17.1
# Deterministic; no random numbers.

import numpy as np
from scipy.integrate import solve_ivp

TWO_PI = 2 * np.pi

def monodromy_trace(G, eps):
    def rhs(tau, y):
        # y = [th1, th1', th2, th2']
        f = G - eps * np.cos(tau)
        return [y[1], f * y[0], y[3], f * y[2]]
    sol = solve_ivp(rhs, (0.0, TWO_PI), [1.0, 0.0, 0.0, 1.0],
                    rtol=1e-13, atol=1e-15, method='DOP853', dense_output=False)
    assert sol.success, sol.message
    th1, dth1, th2, dth2 = sol.y[:, -1]
    det = th1 * dth2 - th2 * dth1 # must be 1: an independent witness
    return th1 + dth2, det

def boundary_G(eps, lo=0.0, hi=None, iters=80):
    """Bisect on G for  $|\text{tr } M| = 2$  at fixed eps. Stable ( $|\text{tr } M| < 2$ ) for small G,
    unstable for large G."""
    if hi is None:
        hi = max(4 * eps**2, 1e-6)
    # make sure the bracket really brackets
    while abs(monodromy_trace(hi, eps)[0]) < 2.0:
        hi *= 2
        if hi > 10:
            raise RuntimeError("no unstable side found")
    if abs(monodromy_trace(lo, eps)[0]) >= 2.0:
        raise RuntimeError("G=0 is already unstable at eps=%g" % eps)
    for _ in range(iters):
        mid = 0.5 * (lo + hi)
        if abs(monodromy_trace(mid, eps)[0]) < 2.0:
            lo = mid
        else:
            hi = mid
    return 0.5 * (lo + hi)

print("CHECK N4 -- exact Floquet stability boundary of the linearised exact")
print("equation, compared with the averaged prediction  $G_c = \text{eps}^{2/2}$ .")
print("No averaging enters the computation; the only approximation is the")
print("linearisation in theta, which is exact for the stability of  $\theta = 0$ .")
print()
print("Sanity: at eps = 0 the equation is  $\theta'' = G\theta$ , so")
print("tr M =  $2\cosh(2\pi\sqrt{G}) > 2$  for every  $G > 0$  -- always unstable.")
for Gtest in [1e-4, 1e-2, 1.0]:
    tr, det = monodromy_trace(Gtest, 0.0)
    print(f"G = {Gtest:.0e}: tr M = {tr:.10f} "
          f" $2\cosh(2\pi\sqrt{G}) = \{2\text{np.cosh}(TWO\_PI\text{np.sqrt}(Gtest))\} \cdot 10^{\text{f}}$  "
          f"det M - 1 = {det-1:.1e}")

```

```

print()

print(f"{'eps=A/l':>9} {'G_c exact':>14} {'eps^2/2':>14} {'ratio':>10} "
      f"{'(ratio-1)/eps^2':>16} {'det M - 1':>11}")
rows = []
for eps in [0.4, 0.2, 0.1, 0.05, 0.025, 0.0125]:
    Gc = boundary_G(eps)
    _, det = monodromy_trace(Gc, eps)
    ratio = Gc / (eps**2 / 2)
    rows.append((eps, Gc, ratio))
    print(f"{'eps:9.5f'} {'Gc:14.9f'} {'eps**2/2:14.9f'} {'ratio:10.6f'} "
          f"{'(ratio-1)/eps**2:16.4f'} {'det-1:+11.1e}")

print()
print("The ratio approaches 1 and (ratio - 1)/eps^2 approaches a constant, so")
print("the exact boundary is G_c = (eps^2/2)*(1 + c*eps^2 + ...): the averaged"
      )
print("condition is the leading term, with a relative error of order eps^2.")
print()

# --- the same statement in physical variables -----
print("Same boundary in physical variables, at g = 9.81 m/s^2, l = 1 m:")
g, l = 9.81, 1.0
print(f"{'A (m)':>9} {'w_c exact':>13} {'sqrt(2gl)/A':>13} {'ratio':>10}")
for eps in [0.4, 0.2, 0.1, 0.05, 0.025, 0.0125]:
    A = eps * l
    Gc = boundary_G(eps)
    wc_exact = np.sqrt(g / (l * Gc))          # G = g/(l w^2)
    wc_claim = np.sqrt(2 * g * l) / A
    print(f"{'A:9.5f'} {'wc_exact:13.4f'} {'wc_claim:13.4f'} {'wc_exact/wc_claim:10.6f'}")

print()
print("MUTATION TEST: which of these candidate boundaries does the exact")
print("Floquet computation actually select, as eps -> 0?")
for name, factor in [("G_c = eps^2/2 (claim)", 0.5),
                     ("G_c = eps^2", 1.0),
                     ("G_c = eps^2/4", 0.25),
                     ("G_c = eps/2", None)]:
    print(f" {name}:")
    for eps in [0.1, 0.025, 0.0125]:
        Gc = boundary_G(eps)
        pred = factor * eps**2 if factor is not None else eps / 2
        print(f"      eps={eps:7.4f}:  G_c(exact)/pred = {Gc/pred:.6f}")

```

check5_basin_and_ripple.py

```

#!/usr/bin/env python3
# check5_basin_and_ripple.py  --  P002, checks N5 and N6
#
# N5.  The effective potential per unit  $m l^2$  is
#       $W(TH) = (g/l) \cos(TH) + (A^2 w^2 / (4 l^2)) \sin^2(TH)$ ,
#      whose stationary points are  $TH = 0$ ,  $TH = \pi$ , and (when  $A^2 w^2 > 2 g l$ )
#       $TH_c = \arccos(2 g l / (A^2 w^2))$ ,
#      a maximum of  $W$  that bounds the well around the inverted position.  The
#      claim tested here is that  $TH_c$  is, to leading order in  $\epsilon = A/l$ , the
#      largest release angle from rest for which the pendulum does not fall.
#      The exact nonlinear equation is integrated and the true boundary found
#      by bisection on the release angle; the two are compared as  $\epsilon \rightarrow 0$ .
#
# N6.  The claim tested is that the fast ripple superposed on the slow motion
#      is  $x_i = (A/l) \sin(TH) \cos(wt)$ , amplitude  $(A/l) \sin(TH)$ .  The ripple

```

```

#      is extracted from the exact solution by subtracting a one-drive-period
#      moving average, and its envelope compared with  $(A/l)*\sin(TH)$ .
#
# Neither check uses any intermediate quantity from the hand derivation other
# than the formula being tested. The trajectories come from the exact
# equation of motion, which check N1 and check N2 test separately.
#
# Environment (pinned): python 3.11.15, numpy 2.4.4, scipy 1.17.1
# Deterministic; no random numbers.

import numpy as np
from scipy.integrate import solve_ivp

g, l = 9.81, 1.0

def exact_rhs(t, s, A, w):
    th, thd = s
    return [thd, (g - A * w**2 * np.cos(w * t)) * np.sin(th) / l]

def falls(th0, A, w, n_slow=25):
    """Integrate from rest at th0 and report whether the pendulum leaves the
    neighbourhood of the inverted position. Escape criterion: |theta| exceeds
    2.5 rad (143 deg), far outside any legitimate slow oscillation about 0 and
    still short of pi, so the test does not hinge on what happens after it
    falls."""
    Om2 = A**2 * w**2 / (2 * l**2) - g / l
    Om = np.sqrt(max(Om2, 1e-12))
    tmax = n_slow * 2 * np.pi / Om
    ev = lambda t, s, A, w: abs(s[0]) - 2.5
    ev.terminal, ev.direction = True, 1.0
    sol = solve_ivp(exact_rhs, (0, tmax), [th0, 0.0], args=(A, w),
                    rtol=1e-9, atol=1e-11, method='DOP853',
                    max_step=2 * np.pi / w / 10, events=ev)
    assert sol.success, sol.message
    return len(sol.t_events[0]) > 0

def boundary_release_angle(A, w, iters=18):
    lo, hi = 1e-3, 1.6
    # lo must survive, hi must fall;
    # hi is above any TH_c tested here
    # and below the escape threshold
    assert not falls(lo, A, w), "even a tiny release angle falls"
    assert falls(hi, A, w), "release at 1.6 rad does not fall"
    for _ in range(iters):
        mid = 0.5 * (lo + hi)
        if falls(mid, A, w):
            hi = mid
        else:
            lo = mid
    return 0.5 * (lo + hi)

print("CHECK N5 -- basin boundary of the inverted position.")
print("TH_c = arccos(2 g l / (A^2 w^2)) is compared with the release angle at")
print("which the exact nonlinear equation first fails to hold the pendulum up."
)
print("The target is held fixed at TH_c = 60 deg while eps = A/l is reduced,")
print("so a discrepancy that does not shrink with eps is a failure.")
print()
TH_C_TARGET = np.deg2rad(60.0)
print("Two comparisons are made. The raw one compares TH_c with the release")

```

```

print("angle theta(0). But releasing FROM REST at theta(0) does not set the")
print("slow amplitude to theta(0): at t = 0 the fast ripple is at its maximum,"
)
print("xi(0) = (A/l) sin(TH), so the slow amplitude is theta(0) - (A/l)sin(TH).")
print("That is an O(eps) offset predicted by the same derivation, so the")
print("corrected column is the sharper test. If the correction were invented")
print("to close a gap it would not also have the right coefficient, which is")
print("why sin(TH_c) = sin(60 deg) = 0.8660 is printed alongside diff/eps.")
print()
print(f"{'eps=A/l':>9} {'A':>8} {'omega':>10} {'TH_c pred':>11} "
      f"{'TH_c exact':>11} {'raw diff':>11} {'diff/eps':>9} {'corrected':>11} "
      f"{'corr/eps^2':>11}")
rows5 = []
for eps in [0.08, 0.04, 0.02, 0.01]:
    A = eps * l
    # solve cos(TH_c) = 2 g l / (A^2 w^2) for w
    w = np.sqrt(2 * g * l / (A**2 * np.cos(TH_C_TARGET)))
    th_pred = np.arccos(2 * g * l / (A**2 * w**2))
    th_num = boundary_release_angle(A, w)
    d = th_num - th_pred
    dc = (th_num - (A / l) * np.sin(th_num)) - th_pred
    rows5.append((eps, d, dc))
    print(f"{eps:9.4f} {A:8.4f} {w:10.2f} {th_pred:11.6f} {th_num:11.6f} "
          f"{d:+11.3e} {d/eps:+9.4f} {dc:+11.3e} {dc/eps**2:+11.4f}")

print()
print(f"sin(TH_c) = {np.sin(TH_C_TARGET):.4f} -- compare the diff/eps column.")
print()
print("Convergence orders:")
for i in range(len(rows5) - 1):
    e0, d0, c0 = rows5[i]
    e1, d1, c1 = rows5[i + 1]
    print(f"eps {e0:.4f} -> {e1:.4f}: order(raw) = "
          f"{np.log(abs(d0/d1))/np.log(e0/e1):5.2f} order(corrected) = "
          f"{np.log(abs(c0/c1))/np.log(e0/e1):5.2f}")

print()
print("Control: is the bisected boundary actually sensitive to the prediction?"
)
print("Below-threshold parameters should give NO stable release angle at all.")
A, l_ = 0.02, 1.0
w_sub = 0.8 * np.sqrt(2 * g * l) / A # below the threshold
try:
    b = boundary_release_angle(A, w_sub)
    print(f"sub-threshold omega = {w_sub:.1f}: boundary found at {b:.4f} rad"
          " <-- unexpected")
except AssertionError as e:
    print(f"sub-threshold omega = {w_sub:.1f}: {e} <-- expected: nothing is"
          "held")

print()
print("=" * 72)
print("CHECK N6 -- amplitude of the fast ripple, claimed to be (A/l)*sin(TH).")
print()
eps = 0.03
A = eps * l
w = 2.0 * np.sqrt(2 * g * l) / A
Om = np.sqrt(A**2 * w**2 / (2 * l**2) - g / l)
Tdrive = 2 * np.pi / w
tmax = 2 * np.pi / Om # one slow period
nper = 400 # samples per drive period
ts = np.linspace(0, tmax, int(tmax / Tdrive) * nper + 1)

```

```

sol = solve_ivp(exact_rhs, (0, tmax), [0.6, 0.0], args=(A, w), t_eval=ts,
                rtol=1e-12, atol=1e-14, method='DOP853', max_step=Tdrive / 40)
assert sol.success
th = sol.y[0]
# slow part: moving average over exactly one drive period
kern = np.ones(nper) / nper
slow = np.convolve(th, kern, mode='same')
ripple = th - slow
# compare the ripple envelope with (A/l)*sin(slow), drive period by period
nblocks = len(th) // nper
print(f"eps = {eps}, omega = {w:.1f}, Omega = {Om:.4f}, "
      f"{nblocks} drive periods in one slow period")
print(f"{'block':>6} {'TH (rad)':>10} {'ripple amp':>12} {'(A/l)|sinTH|':>13} "
      f"{'ratio':>9}")
ratios = []
for b in range(2, nblocks - 2):
    # skip the edges of the convolution
    sl = slice(b * nper, (b + 1) * nper)
    amp = 0.5 * (ripple[sl].max() - ripple[sl].min())
    THb = slow[sl].mean()
    pred = (A / l) * abs(np.sin(THb))
    if b % max(1, (nblocks // 10)) == 0:
        print(f"{b:6d} {THb:10.5f} {amp:12.5e} {pred:13.5e} {amp/pred:9.5f}")
    ratios.append(amp / pred)
ratios = np.array(ratios)
print()
print(f"ratio over all {len(ratios)} interior drive periods: "
      f"mean {ratios.mean():.5f}, min {ratios.min():.5f}, max {ratios.max():.5f} ")
print(f"Expected 1 + O(eps) = 1 +/- {eps:.2f}. "
      f"{'PASS' if abs(ratios.mean()-1) < 3*eps else 'FAIL'}")
print()
print("The ratio is the wrong statistic near TH = 0, where the predicted")
print("amplitude (A/l)|sin TH| passes through zero and the O(eps^2) remainder")
print("dominates it; the outliers above are exactly those blocks. Two better")
print("conditioned statistics follow. They are additional diagnostics, not a")
print("replacement of the criterion above, which was met as stated.")
sin_all, amp_all, pred_all = [], [], []
for b in range(2, nblocks - 2):
    sl = slice(b * nper, (b + 1) * nper)
    amp_all.append(0.5 * (ripple[sl].max() - ripple[sl].min()))
    THb = slow[sl].mean()
    sin_all.append(abs(np.sin(THb)))
    pred_all.append((A / l) * abs(np.sin(THb)))
sin_all = np.array(sin_all); amp_all = np.array(amp_all); pred_all = np.array(
    pred_all)
sel = sin_all > 0.2
rsel = amp_all[sel] / pred_all[sel]
print(f" (i) ratio restricted to blocks with |sin TH| > 0.2 "
      f"({sel.sum()} blocks): mean {rsel.mean():.5f}, "
      f"min {rsel.min():.5f}, max {rsel.max():.5f}")
absres = np.abs(amp_all - pred_all)
print(f" (ii) max absolute residual |amp - (A/l)|sin TH|| = {absres.max():.3e} "
      f" = {absres.max()/eps**2:.3f} * eps^2 (expected O(eps^2))")
print()
print("Mutation: the same ratio against a constant-amplitude ripple A/l")
print("(i.e. dropping the sin(TH) factor):")
pred_const = A / l
r2 = []
for b in range(2, nblocks - 2):
    sl = slice(b * nper, (b + 1) * nper)
    r2.append(0.5 * (ripple[sl].max() - ripple[sl].min()) / pred_const)
r2 = np.array(r2)

```

```
print(f"    mean {r2.mean():.5f}, min {r2.min():.5f}, max {r2.max():.5f} "
      f"-> {'REJECTED' if r2.min() < 0.9 else 'not rejected'}")
```

check6_duffing_shift.py

```
#!/usr/bin/env python3
# check6_duffing_shift.py -- P002, check N7 (new in round 2)
#
# Purpose: test the amplitude-frequency relation derived by Lindstedt-Poincare
# in §3.9 of the answer,
#
#  $x'' + \Omega^2 x + \alpha x^3 = 0 \Rightarrow \omega(a) = \Omega (1 + 3 \alpha a^2 / (8 \Omega^2))$ 
# and hence  $T(a)/T(0) - 1 = -3 \alpha a^2 / (8 \Omega^2) + O(a^4)$ . (claim)
#
# In round 1 this relation was recalled rather than derived, tagged [R], and
# tested only indirectly (as column (c) of check N3, which measured it inside
# the pendulum problem). Finding F1.2 of audit 1 asked for the provenance to
# be
# fixed. It is now derived in the text, and this script tests the derived
# relation directly on the bare Duffing equation, away from the pendulum, so
# that the two are independent.
#
# Design: the residual is  $O(a^4)$  relative, so a single tolerance proves little.
# The amplitude is swept down by a factor of 8 and the residual is required to
# fall like  $a^2$  RELATIVE to the shift itself (equivalently  $a^4$  absolute), which
# a wrong coefficient cannot fake. Mutants with  $3/8$  replaced by  $1/8$ ,  $3/4$  and
#  $-3/8$  are run alongside.
#
# Environment (pinned): python 3.11.15, numpy 2.4.4, scipy 1.17.1
# Deterministic; no random numbers.

import numpy as np
from scipy.integrate import solve_ivp

OM = 6.0 # same Omega as check N3 uses, for comparability
G_OVER_L = 9.81 # only used to build the pendulum's own alpha

def duffing_rhs(t, s, Om, alpha):
    x, v = s
    return [v, -Om**2 * x - alpha * x**3]

def measured_period(a, Om, alpha, n_periods=40):
    """Period from a straight-line fit to the zero-crossing times."""
    tmax = n_periods * 2 * np.pi / Om
    ts = np.linspace(0.0, tmax, 200001)
    sol = solve_ivp(duffing_rhs, (0, tmax), [a, 0.0], t_eval=ts,
                    args=(Om, alpha), rtol=1e-12, atol=1e-14, method='DOP853')
    assert sol.success, sol.message
    x = sol.y[0]
    sg = np.sign(x)
    idx = np.where(sg[:-1] * sg[1:] < 0)[0]
    t0, t1, y0, y1 = ts[idx], ts[idx + 1], x[idx], x[idx + 1]
    tc = t0 - y0 * (t1 - t0) / (y1 - y0)
    k = np.arange(len(tc))
    slope, _ = np.polyfit(k, tc, 1)
    return 2 * slope, len(tc)
```

```

print("CHECK N7 -- amplitude-frequency relation for  $x'' + \Omega^2 x + \alpha x^3 = 0$ ." )
print("Derived in §3.9 by Lindstedt-Poincare; tested here on the bare Duffing")
print("equation, with no pendulum anywhere in the script.")
print()

# alpha chosen to be the pendulum's own: the effective well of §3.8 has
#  $T_{Hdd} = -\Omega^2 T_H - 4 \beta T_H^3$  with  $\beta = -(\Omega^2/6 + (g/l)/8)$ , so  $\alpha = 4 \beta$ .
beta = -(\Omega**2 / 6 + G_OVER_L / 8)
alpha = 4 * beta
print(f" $\Omega = \{\Omega\}$ ,  $\beta = \{\beta:.6f\}$ ,  $\alpha = 4\beta = \{\alpha:.6f\}$ ")
print("T(0) is taken as  $2\pi/\Omega$ , the exact  $a \rightarrow 0$  limit of the Duffing period."
)
print()

T0 = 2 * np.pi / Ω
print(f"{'a':>9} {'T_meas':>12} {'meas shift':>13} {'pred shift':>13} "
      f"{'residual':>12} {'resid/a^4':>11} {'ncross':>7}")
rows = []
for a in [0.16, 0.08, 0.04, 0.02]:
    T, nc = measured_period(a, Ω, alpha)
    meas = T / T0 - 1
    pred = -3 * alpha * a**2 / (8 * Ω**2)
    res = meas - pred
    rows.append((a, meas, pred, res))
    print(f"{'a':9.4f} {'T':12.8f} {'meas':+13.5e} {'pred':+13.5e} {'res':+12.3e} "
          f"{'res/a**4':11.4f} {'nc':7d}")

print()
print("Convergence order of the residual (expected 4: the next term is  $O(a^4)$ ):")
)
for i in range(len(rows) - 1):
    a0, _, _, r0 = rows[i]
    a1, _, _, r1 = rows[i + 1]
    print(f"  a {'a0':.4f} -> {'a1':.4f}:  order = "
          f"{'np.log(abs(r0/r1))/np.log(a0/a1):5.2f}")

print()
print("Cross-check against check N3.  At  $a = 0.02$ , the shift predicted here is")
)
print("the eps-independent floor that column (c) of N3 measured inside the")
print("pendulum problem.")
a = 0.02
pred = -3 * alpha * a**2 / (8 * Ω**2)
print(f"  predicted period shift at  $a = \{a\}$ : {'pred':+.5e}")
print(f"  N3 column (c), measured in the pendulum:  +1.205e-04")
print(f"  (N3's column (c) is a PERIOD shift, so it should equal this value.)")

print()
print("MUTATION TEST: same measurement against wrong coefficients.")
for name, coef in [(" $3/8$  (claim)", 3 / 8), (" $1/8$ ", 1 / 8), (" $3/4$ ", 3 / 4),
                  (" $-3/8$  (sign flipped)", -3 / 8)]:
    a = 0.08
    T, _ = measured_period(a, Ω, alpha)
    meas = T / T0 - 1
    pred = -coef * alpha * a**2 / Ω**2
    rel = abs(meas - pred) / abs(meas)
    print(f"  coefficient {name:22s}: predicted {'pred':+.5e}, measured "
          f"{'meas':+.5e}, rel err {'rel':8.4f} -> "
          f"{'consistent' if rel < 0.02 else 'REJECTED'}")

```

check7_floquet_frequency.py

```
#!/usr/bin/env python3
# check7_floquet_frequency.py -- P002, check N8 (new in round 2)
#
# Purpose: test the closed form
#
#  $\Omega = \sqrt{A^2 w^2 / (2 l^2) - g / l}$  (claim)
#
# by a route that contains no averaging and no time-series fitting: the Floquet
# phase of the exact linearised equation. This check was suggested by the
# round-1 auditor, who ran a single-point version of it in audit pass 6; it is
# adopted here, swept in eps, and credited in S8.
#
# Because  $\theta = 0$  is an exact solution of the exact equation of motion, the
# linearisation about it is exact for the question of small oscillations. With
#  $\tau = w t$ ,
#
#  $\theta'' = (G - \epsilon \cos \tau) \theta$ ,  $G = g / (l w^2)$ ,  $\epsilon = A / l$ ,
#
# a Hill equation of period  $2\pi$ . Its monodromy matrix  $M$  satisfies  $\det M = 1$ 
# (derived in S4.4), so inside the stable band the multipliers are  $\exp(\pm i \nu)$ 
# with  $2 \cos \nu = \text{tr } M$ .  $\nu$  is the phase the slow motion advances per DRIVE
# period, so the slow angular frequency is
#
#  $\Omega_{\text{exact}} = \nu * w / (2\pi)$ ,  $\nu = \arccos(\text{tr } M / 2)$ .
#
# Nothing here fits a period, and nothing averages. N3 measures a period from
# the full NONLINEAR equation at finite amplitude; this measures the exact
# linear-response frequency. The two are independent and fail differently.
#
# Environment (pinned): python 3.11.15, numpy 2.4.4, scipy 1.17.1
# Deterministic; no random numbers.

import numpy as np
from scipy.integrate import solve_ivp

TWO_PI = 2 * np.pi
g, l = 9.81, 1.0

def monodromy(G, eps, rtol=1e-13, atol=1e-15):
    def rhs(tau, y):
        f = G - eps * np.cos(tau)
        return [y[1], f * y[0], y[3], f * y[2]]
    s = solve_ivp(rhs, (0.0, TWO_PI), [1.0, 0.0, 0.0, 1.0],
                  rtol=rtol, atol=atol, method='DOP853')
    assert s.success, s.message
    th1, dth1, th2, dth2 = s.y[:, -1]
    return th1 + dth2, th1 * dth2 - th2 * dth1

def omega_floquet(A, w):
    G = g / (l * w**2)
    tr, det = monodromy(G, A / l)
    if abs(tr) >= 2.0:
        return np.nan, tr, det
    nu = np.arccos(tr / 2.0)
    return nu * w / TWO_PI, tr, det

def omega_claim(A, w):
    v = A**2 * w**2 / (2 * l**2) - g / l
```

```

    return np.sqrt(v) if v > 0 else np.nan

print("CHECK N8 -- slow frequency from the Floquet phase of the exact")
print("linearised equation. No averaging, no period fitting.")
print()
print("First, reproduction of the round-1 auditor's single-point computation")
print("(their pass 6): l = 1 m, g = 9.81, eps = 0.05, omega = 150 s-1.")
Om_f, tr, det = omega_floquet(0.05 * l, 150.0)
print(f"    tr M = {tr:.12f}    det M - 1 = {det-1:+.2e}")
print(f"    Omega from Floquet phase = {Om_f:.4f}")
print(f"    Omega from the claim      = {omega_claim(0.05*l, 150.0):.4f}")
print("    auditor reported          Exact: 4.2764, Claimed: 4.2796")
print()

print("Now the sweep. Omega is held at 6.000 s-1 while eps falls by a factor")
print("of 16; omega is solved for from the claim at each eps. If the claim is")
print("right the Floquet frequency converges on it like eps2. A residual that")
print("plateaus, or falls at the wrong order, is a failure.")
print()
OM = 6.0
print(f"{'eps=A/l':>9} {'omega':>10} {'G':>12} {'tr M':>14} "
      f"{'Omega_floquet':>14} {'rel err':>12} {'err/eps^2':>10} {'det M - "
      f"1':>11}")
rows = []
for eps in [0.1, 0.05, 0.025, 0.0125, 0.00625]:
    A = eps * l
    w = (np.sqrt(2.0) * l / A) * np.sqrt(OM**2 + g / l)
    Om_f, tr, det = omega_floquet(A, w)
    rel = Om_f / OM - 1.0
    rows.append((eps, rel))
    print(f"{eps:9.5f} {w:10.2f} {g/(l*w**2):12.3e} {tr:14.10f} "
          f"{Om_f:14.9f} {rel:+12.3e} {rel/eps**2:+10.4f} {det-1:+11.1e}")

print()
print("Convergence order (log-log slopes between consecutive eps):")
for i in range(len(rows) - 1):
    e0, r0 = rows[i]
    e1, r1 = rows[i + 1]
    print(f"    eps {e0:.5f} -> {e1:.5f}: order = "
          f"{np.log(abs(r0/r1))/np.log(e0/e1):5.2f}")

print()
print("Independence from check N3. N3 fits a period from the full nonlinear")
print("equation at finite amplitude and had to subtract an anharmonic shift;")
print("this check linearises and reads a Floquet phase, with no fitting and no")
print("anharmonicity. The two coefficients below are therefore different")
print("numbers measuring different things, and both must be O(1).")
print(f"    N8 (this check), rel err / eps^2 -> {rows[-1][1]/rows[-1][0]**2:+.4f}")
print("    N3 column (b), rel err / eps^2 -> -0.758 (period, nonlinear)")

print()
print("=" * 78)
print("CONSISTENCY BETWEEN N4 AND N8. N4 found the exact stability boundary to")
print("sit at a HIGHER drive frequency than sqrt(2gl)/A, so just above the")
print("averaged threshold the exact system must still be unstable while the")

```

```

print("closed form already predicts a positive Omega. The O(eps^2) correction"
)
print("to Omega must therefore be NEGATIVE near threshold, even though the")
print("sweep above found it positive far from threshold. If it does not change
")
print("sign, N4 and N8 contradict each other and one of them is wrong.")
print()
print("eps is held at 0.1 and omega/omega_c is scanned. mu = 2gl/(A^2 w^2)")
print("= cos(Theta_c) measures nearness to threshold: mu = 1 at threshold,")
print("mu -> 0 far above it.")
print()
eps = 0.1
A = eps * l
wc_claim = np.sqrt(2 * g * l) / A
print(f"{'w/w_c':>8} {'mu':>8} {'Omega_claim':>12} {'Omega_floq':>12} "
      f"{'difference':>12} {'rel':>11}")
for r in [1.0, 1.002, 1.01, 1.03, 1.1, 1.3, 1.8, 2.5, 3.5]:
    w = r * wc_claim
    mu = 2 * g * l / (A**2 * w**2)
    Om_c = omega_claim(A, w)
    Om_f, tr, _ = omega_floquet(A, w)
    d = Om_f - Om_c
    rel = d / Om_c if (Om_c and Om_c > 0) else np.nan
    fo = f"{Om_f:12.6f}" if np.isfinite(Om_f) else f"{'unstable':>12}"
    dd = f"{d:+12.6f}" if np.isfinite(d) else f"{'--':>12}"
    rr = f"{rel:+11.3e}" if np.isfinite(rel) else f"{'--':>11}"
    print(f"{r:8.3f} {mu:8.4f} {Om_c:12.6f} {fo} {dd} {rr}")
print()
Gc = None
lo, hi = 0.0, 4 * eps**2
for _ in range(70):
    mid = 0.5 * (lo + hi)
    lo, hi = (mid, hi) if abs(monodromy(mid, eps)[0]) < 2.0 else (lo, mid)
Gc = 0.5 * (lo + hi)
wc_exact = np.sqrt(g / (l * Gc))
print(f"exact boundary from the N4 bisection: omega_c = {wc_exact:.4f} "
      f"= {wc_exact/wc_claim:.6f} * sqrt(2gl)/A")
print("The table's first rows show Omega_floquet undefined (unstable) exactly")
print("where omega < that value, and the difference entering negative just")
print("above it, then crossing zero and turning positive. N4 and N8 agree.")

print()
print("MUTATION TEST: the Floquet frequency against wrong closed forms, at")
print("eps = 0.02. omega is solved for from each candidate so that all four")
print("would predict Omega = 6.000 if they were right.")
eps = 0.02
A = eps * l
cands = {
    "claim    Omega^2 = A^2w^2/(2l^2) - g/l": np.sqrt(2.0),
    "mutant   Omega^2 = A^2w^2/(1l^2) - g/l": 1.0,
    "mutant   Omega^2 = A^2w^2/(4l^2) - g/l": 2.0,
}
for name, c in cands.items():
    w = (c * l / A) * np.sqrt(OM**2 + g / l)
    Om_f, tr, _ = omega_floquet(A, w)
    rel = abs(Om_f / OM - 1.0)
    print(f" {name}:  omega = {w:9.2f}  Omega_floquet = {Om_f:9.5f} "
          f"rel err = {rel:9.3e} -> {'consistent' if rel < 1e-2 else 'REJECTED'
          }")
w = (np.sqrt(2.0) * l / A) * np.sqrt(max(OM**2 - g / l, 1e-9))
Om_f, tr, _ = omega_floquet(A, w)
rel = abs(Om_f / OM - 1.0)
print(f" mutant   Omega^2 = A^2w^2/(2l^2) + g/l:  omega = {w:9.2f} ")

```

```
f"Omega_floquet = {Om_f:9.5f} rel err = {rel:9.3e} -> "
f"{'consistent' if rel < 1e-2 else 'REJECTED'}")
```

figures.py

```
#!/usr/bin/env python3
# figures.py -- P002, figures 1-3
#
# Fig 1 fig1-slow-oscillation.png
# (a) theta(t) from the exact equation over three slow periods, with the
# averaged (slow) solution over it;
# (b) a zoom of three drive periods showing the fast ripple, with the
# predicted  $TH + (A/l) \sin(TH) \cos(w t)$  over it.
# Fig 2 fig2-stability.png
# (a) where the inverted position is held, in the  $(A/l, \omega)$  plane:
# direct integration of the exact nonlinear equation (markers), the
# exact Floquet boundary of the linearised equation (solid), and the
# averaged prediction  $A^2 \omega^2 = 2 g l$  (dashed);
# (b) theta(t) at three points across the boundary.
# Fig 3 fig3-effective-potential.png
# the effective potential  $W(TH)$  below, at, and above threshold, with the
# barrier angle  $TH_c$  marked.
#
# Environment (pinned): python 3.11.15, numpy 2.4.4, scipy 1.17.1,
# matplotlib 3.10.9. Deterministic; no random numbers.

import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

g, l = 9.81, 1.0
OUT = "."

plt.rcParams.update({
    "font.size": 9, "axes.titlesize": 9.5, "axes.labelsize": 9,
    "figure.dpi": 160, "savefig.dpi": 160, "savefig.bbox": "tight",
    "axes.grid": True, "grid.alpha": 0.25, "grid.linewidth": 0.5,
    "axes.spines.top": False, "axes.spines.right": False,
    "legend.frameon": False, "legend.fontsize": 8,
})

C_EXACT, C_AVG, C_PRED = "#1b3a6b", "#c8102e", "#0f8a5f"

def exact_rhs(t, s, A, w):
    th, thd = s
    return [thd, (g - A * w**2 * np.cos(w * t)) * np.sin(th) / l]

def averaged_rhs(t, s, A, w):
    TH, THd = s
    return [THd, (g / l) * np.sin(TH) - (A**2 * w**2 / (2 * l**2)) * np.sin(TH) * np.cos(TH)]

def Omega(A, w):
    v = A**2 * w**2 / (2 * l**2) - g / l
    return np.sqrt(v) if v > 0 else np.nan
```

```

# ===== figure 1 =====
A, eps = 0.05, 0.05
w = 2.0 * np.sqrt(2 * g * l) / A          # twice the threshold frequency
Om = Omega(A, w)
Tslow, Tdrive = 2 * np.pi / Om, 2 * np.pi / w
th0 = 0.35

tmax = 3 * Tslow
ts = np.linspace(0, tmax, 200001)
ex = solve_ivp(exact_rhs, (0, tmax), [th0, 0.0], t_eval=ts, args=(A, w),
               rtol=1e-11, atol=1e-13, method='DOP853', max_step=Tdrive / 30)
# the averaged solution is started from the SLOW part of the same initial
# state: theta(0) = TH(0) + (A/l) sin TH(0) at t = 0, so TH(0) solves
# TH + (A/l) sin TH = th0.
TH0 = th0
for _ in range(60):
    TH0 -= (TH0 + (A / l) * np.sin(TH0) - th0) / (1 + (A / l) * np.cos(TH0))
av = solve_ivp(averaged_rhs, (0, tmax), [TH0, 0.0], t_eval=ts, args=(A, w),
               rtol=1e-11, atol=1e-13, method='DOP853')
assert ex.success and av.success

fig, ax = plt.subplots(3, 1, figsize=(6.6, 7.2))
ax[0].plot(ts, np.rad2deg(ex.y[0]), lw=0.6, color=C_EXACT,
           label=r"exact $\theta(t)$")
ax[0].plot(ts, np.rad2deg(av.y[0]), lw=1.6, color=C_AVG, ls="--",
           label=r"averaged $\Theta(t)$")
ax[0].axhline(0, color="k", lw=0.6, alpha=0.5)
ax[0].set_xlabel("t [s]")
ax[0].set_ylabel(r"$\theta$ [deg]")
ax[0].set_title(
    rf"$l={l:g}$ m, $A={A:g}$ m ($A/l={eps:g}$), $\omega={w:.1f}$\,,$s^{\{-1\}}$"
    rf"$=2\omega_{\{\rm c\}}$" "\n"
    rf"$\Omega={Om:.3f}$\,,$s^{\{-1\}}$, released from "
    rf"$\theta(0)={np.rad2deg(th0):.1f}$\circ$ at rest")
ax[0].legend(loc="lower center", ncol=2)
ax[0].set_xlim(0, tmax)
ax[0].set_ylim(-34, 30)

# zoom: three drive periods starting at t = 0, where the slow amplitude is at
# its maximum and the ripple is therefore largest
m = ts <= 3 * Tdrive
THz = np.interp(ts[m], ts, av.y[0])
pred = THz + (A / l) * np.sin(THz) * np.cos(w * ts[m])
ax[1].plot(ts[m] * 1e3, np.rad2deg(ex.y[0][m]), lw=1.4, color=C_EXACT,
           label=r"exact $\theta(t)$")
ax[1].plot(ts[m] * 1e3, np.rad2deg(THz), lw=1.4, ls="--", color=C_AVG,
           label=r"averaged $\Theta(t)$")
ax[1].plot(ts[m] * 1e3, np.rad2deg(pred), lw=1.2, ls=":", color=C_PRED,
           label=r"$\Theta+(A/l)\sin\Theta$, $\cos\omega t$")
ax[1].set_xlabel("t [ms]")
ax[1].set_ylabel(r"$\theta$ [deg]")
ax[1].set_title("three drive periods at the turning point: the fast ripple")
ax[1].legend(loc="lower left")

# residual over one slow period, against the predicted envelope
m2 = ts <= Tslow
res = ex.y[0][m2] - np.interp(ts[m2], ts, av.y[0])
env = (A / l) * np.abs(np.sin(np.interp(ts[m2], ts, av.y[0])))
ax[2].plot(ts[m2], np.rad2deg(res), lw=0.5, color=C_EXACT,
           label=r"$\theta(t)-\Theta(t)$")
ax[2].plot(ts[m2], np.rad2deg(env), lw=1.3, color=C_PRED,
           label=r"$\pm (A/l)$, $|\sin\Theta(t)|$")
ax[2].plot(ts[m2], -np.rad2deg(env), lw=1.3, color=C_PRED)

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ax[2].set_xlabel("t [s]")
ax[2].set_ylabel(r"$\theta$-\Theta$ [deg]")
ax[2].set_title("the ripple over one slow period, against its predicted
    envelope")
ax[2].legend(loc="lower center", ncol=2)
ax[2].set_xlim(0, Tslow)
fig.tight_layout()
fig.savefig(f"{OUT}/fig1-slow-oscillation.png")
plt.close(fig)
print("fig1 written; Omega =", Om, " Tslow =", Tslow, " Tdrive =", Tdrive)

# ===== figure 2 =====
TWO_PI = 2 * np.pi

def floquet_trace(G, e):
    def rhs(tau, y):
        f = G - e * np.cos(tau)
        return [y[1], f * y[0], y[3], f * y[2]]
    s = solve_ivp(rhs, (0, TWO_PI), [1.0, 0.0, 0.0, 1.0],
        rtol=1e-12, atol=1e-14, method='DOP853')
    return s.y[0, -1] + s.y[3, -1]

def floquet_Gc(e):
    lo, hi = 0.0, max(4 * e**2, 1e-8)
    while abs(floquet_trace(hi, e)) < 2.0:
        hi *= 2
    for _ in range(60):
        mid = 0.5 * (lo + hi)
        if abs(floquet_trace(mid, e)) < 2.0:
            lo = mid
        else:
            hi = mid
    return 0.5 * (lo + hi)

def survives(A, w, th0=0.1, tmax=6.0):
    ev = lambda t, s, A, w: abs(s[0]) - 2.5
    ev.terminal, ev.direction = True, 1.0
    s = solve_ivp(exact_rhs, (0, tmax), [th0, 0.0], args=(A, w), rtol=1e-9,
        atol=1e-11, method='DOP853', max_step=2 * np.pi / w / 10,
        events=ev)
    return len(s.t_events[0]) == 0

eps_grid = np.linspace(0.02, 0.30, 22)
w_ratio = np.linspace(0.5, 1.6, 22)
pts_ok, pts_no = [], []
for e in eps_grid:
    Aa = e * l
    wc = np.sqrt(2 * g * l) / Aa
    for r in w_ratio:
        (pts_ok if survives(Aa, r * wc) else pts_no).append((e, r))
pts_ok, pts_no = np.array(pts_ok), np.array(pts_no)

eps_fine = np.linspace(0.02, 0.30, 40)
r_floq = np.array([np.sqrt(g / (l * floquet_Gc(e))) / (np.sqrt(2 * g * l) / (e
    * l))
    for e in eps_fine])

fig, ax = plt.subplots(1, 2, figsize=(9.2, 3.6),
    gridspec_kw=dict(width_ratios=[1.15, 1]))

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ax[0].scatter(pts_ok[:, 0], pts_ok[:, 1], s=9, c=C_EXACT, marker="o",
              label="stays up (nonlinear,  $\theta_0=0.1$  rad)")
ax[0].scatter(pts_no[:, 0], pts_no[:, 1], s=9, c="#b0b0b0", marker="x",
              label="falls over")
ax[0].plot(eps_fine, r_floq, color=C_PRED, lw=1.6,
           label="exact Floquet boundary")
ax[0].axhline(1.0, color=C_AVG, ls="--", lw=1.6,
              label=r"averaged:  $A^2\omega^2=2gl$ ")
ax[0].set_xlabel(r" $\epsilon = A/l$ ")
ax[0].set_ylabel(r" $\omega, \omega_c$ ",  $\omega_c = \sqrt{2gl}/A$ )
ax[0].set_title("where the inverted position is held")
ax[0].legend(loc="upper center", bbox_to_anchor=(0.5, -0.22), ncol=2)
ax[0].set_ylim(w_ratio[0], w_ratio[-1])

A2 = 0.05
wc2 = np.sqrt(2 * g * l) / A2
for r, ls, lab in [(1.4, "-", r" $\omega=1.4, \omega_c$ : held"),
                  (1.02, "-", r" $\omega=1.02, \omega_c$ : marginal"),
                  (0.9, "-", r" $\omega=0.9, \omega_c$ : falls")]:
    ww = r * wc2
    tm = 4.0
    tt = np.linspace(0, tm, 40001)
    s = solve_ivp(exact_rhs, (0, tm), [0.1, 0.0], t_eval=tt, args=(A2, ww),
                  rtol=1e-10, atol=1e-12, method='DOP853',
                  max_step=2 * np.pi / ww / 20)
    ax[1].plot(tt, np.rad2deg(s.y[0]), lw=0.9, ls=ls, label=lab)
ax[1].set_xlabel("t [s]")
ax[1].set_ylabel(r" $\theta$  [deg]")
ax[1].set_title(rf" $A/l={A2:g}$ , released from  $\theta_0=5.7^\circ$  at rest")
ax[1].legend(loc="upper center", bbox_to_anchor=(0.5, -0.22), ncol=1)
ax[1].set_ylim(-40, 200)
fig.tight_layout()
fig.savefig(f"{OUT}/fig2-stability.png")
plt.close(fig)
print("fig2 written; n stable =", len(pts_ok), " n fallen =", len(pts_no))

# ===== figure 3 =====
TH = np.linspace(-np.pi, np.pi, 2401)
fig, axs = plt.subplots(1, 2, figsize=(8.6, 3.4),
                        gridspec_kw=dict(width_ratios=[1.5, 1]))

ax, axin = axs
A3 = 0.05
wc3 = np.sqrt(2 * g * l) / A3
for r, c in [(0.7, "#9aa3ad"), (1.0, "#6b7280"), (1.5, C_EXACT), (2.2, C_PRED)]:
    ww = r * wc3
    W = (g / l) * np.cos(TH) + (A3**2 * ww**2 / (4 * l**2)) * np.sin(TH)**2
    W0 = (g / l) # W at Theta = 0, exactly
    for a_, lw_ in ((ax, 1.4), (axin, 1.4)):
        a_.plot(np.rad2deg(TH), W - W0, color=c, lw=lw_,
                label=(rf" $\omega={r:g}, \omega_c$ " if a_ is ax else
                       None))
    q = 2 * g * l / (A3**2 * ww**2)
    if q < 1:
        thc = np.arccos(q)
        Wc = (g / l) * np.cos(thc) + (A3**2 * ww**2 / (4 * l**2)) * np.sin(thc)**2
        ax.plot([np.rad2deg(thc), -np.rad2deg(thc)], [Wc - W0] * 2, "o",
                ms=4.5, color=c)
for a_ in (ax, axin):
    a_.axvline(0, color="k", lw=0.6, alpha=0.4)
    a_.set_xlabel(r" $\theta$  [deg from upward vertical]")

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    a_.set_ylabel(r"$W(\Theta)-W(0)$ [s$^{-2}$]")
ax.set_title(r"effective potential per unit $ml^2$; dots mark $\Theta_{\rm c}$")
)
ax.legend(loc="lower center", ncol=4, bbox_to_anchor=(0.5, -0.46))
axin.set_xlim(-30, 30)
axin.set_ylim(-0.8, 1.6)
axin.axhline(0, color="k", lw=0.5, alpha=0.4)
axin.set_title(r"detail near $\Theta=0$: the well that holds it up")
fig.tight_layout()
fig.savefig(f"{OUT}/fig3-effective-potential.png")
plt.close(fig)
print("fig3 written")

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